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Chapter 8-2
Translation out of SSA

Recap: Dominance

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- A **dom** B
  - if all paths from Entry to B goes through A
- A **post-dom** B
  - if all paths from B to Exit goes through A

Recap: Dominance

- A **dom** B
  - if all paths from Entry to B goes through A
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- Strict (post-)dominance – A (post-)**dom** B but A ≠ B
- Immediate dominance – A strict-dom B, but there's no C, such that A strict-dom C, C strict-dom B

Recap: Dominance Frontier

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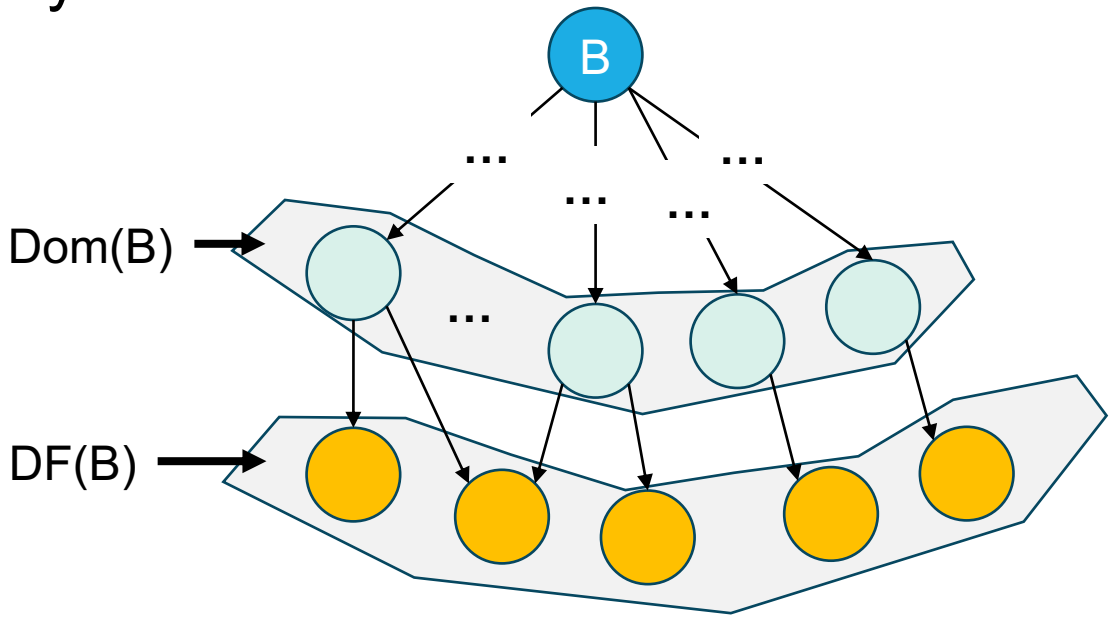
- DF(B) = { … } for the block B
  - The immediate successors of the blocks dominated by B
  - Not strictly dominated by B



Recap: Dominance Frontier

- DF(B) = { … } for the block B
  - The immediate successors of the blocks dominated by B
  - Not strictly dominated by B

- DF($\mathbb{B}$) = { … } for a set of blocks $\mathbb{B}$
  - DF($\mathbb{B}$) = $\cup_{B \in \mathbb{B}}$ DF(B)

Recap: Iterated Dominance Frontier

- Iterated DF of a block set $\mathbb{B}$

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Recap: Iterated Dominance Frontier

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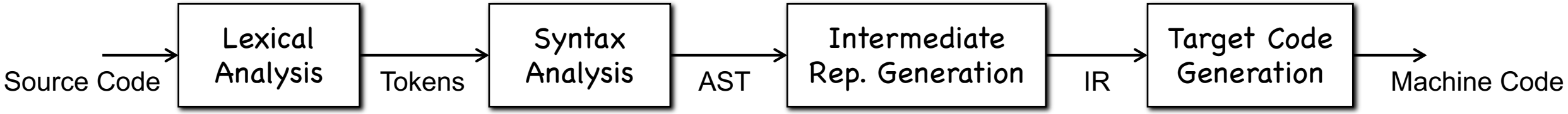
  - ……
  - until fixed point! (i.e., $DF_n = DF_{n-1}$)

Recap: Why SSA?



Recap: Why SSA? What are the two features of SSA?

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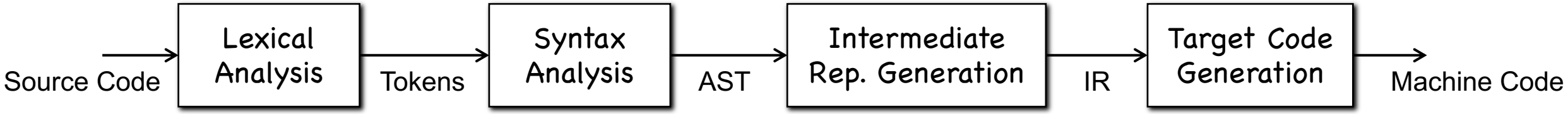
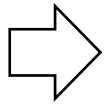


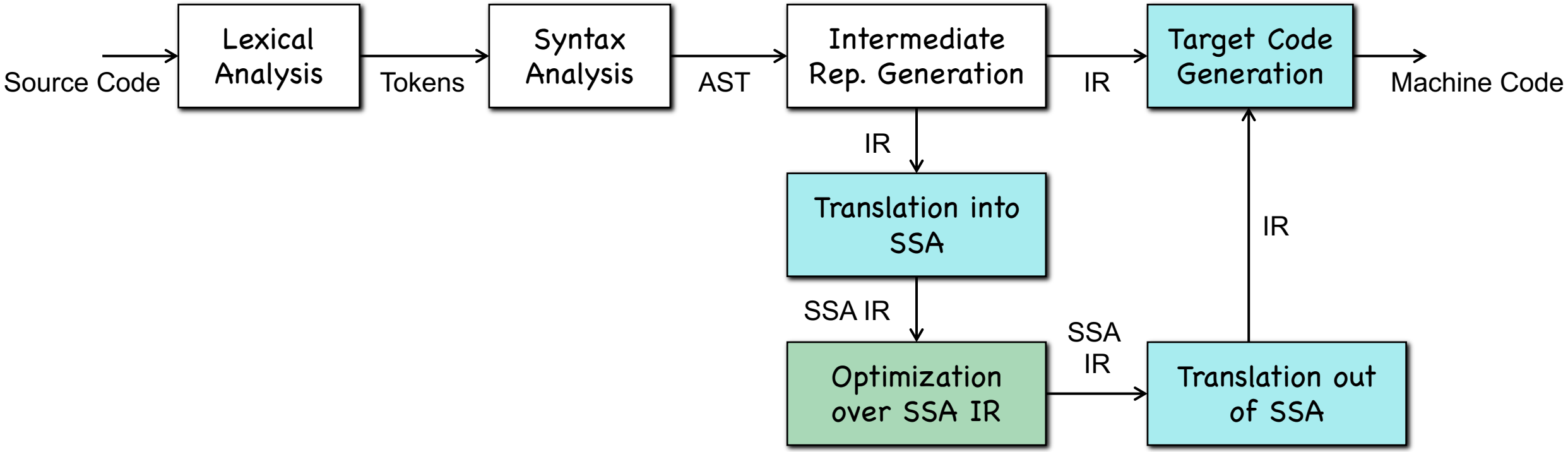
Recap: Why SSA? What are the two features of SSA?

```
1. a = b + c
2. b = a - d
3. c = b + c
4. d = a - d
```

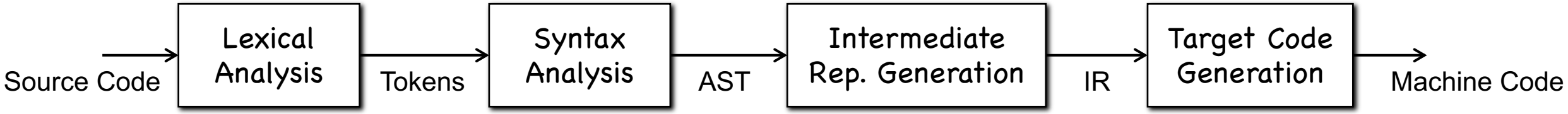


```
1. a1 = b1 + c1
2. b2 = a1 - d1
3. c2 = b2 + c1
4. d1 = a1 - d1
```

Recap: SSA in Compilers



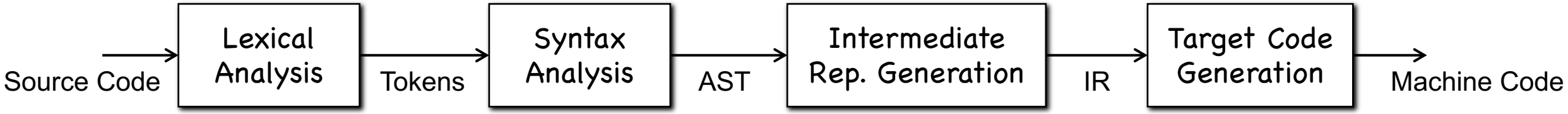
Recap: Translation into SSA



Recap: How can we generate IR in SSA form?

Hints: Two steps corresponding to two features of SSA.

Recap: Translation into SSA



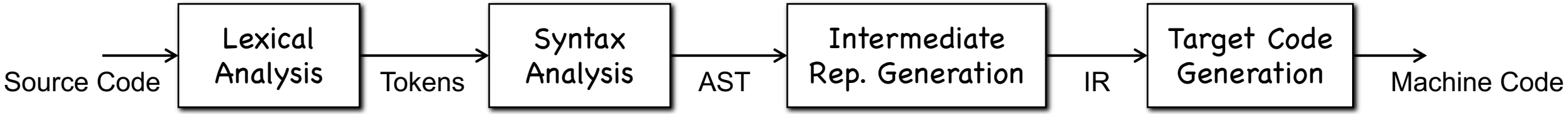
Recap: How can we generate IR in SSA form?

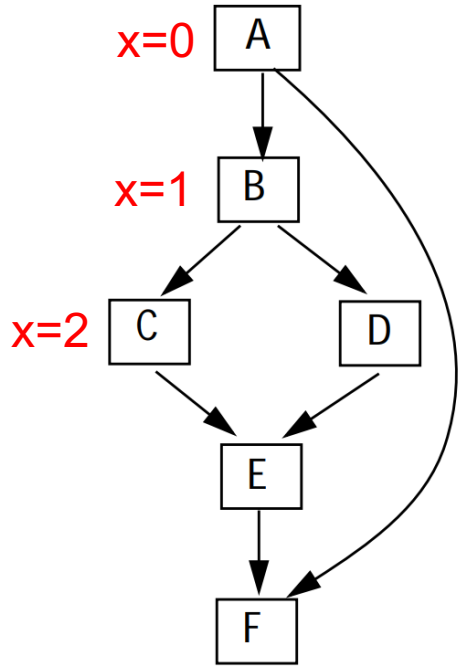
Hints: Two steps corresponding to two features of SSA.

Step 1: Inserting **ϕ** at the iterated dominance frontier to merge definitions.

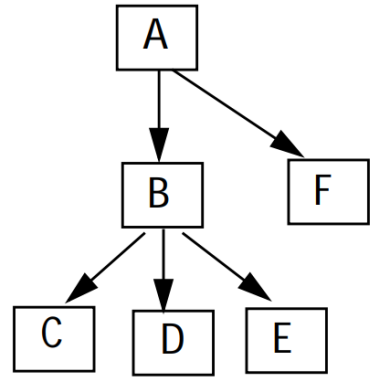
Step 2: Using subscripts to distinguish definitions of the same variable.

Recap: Translation into SSA





Control Flow Graph



Dominator Tree

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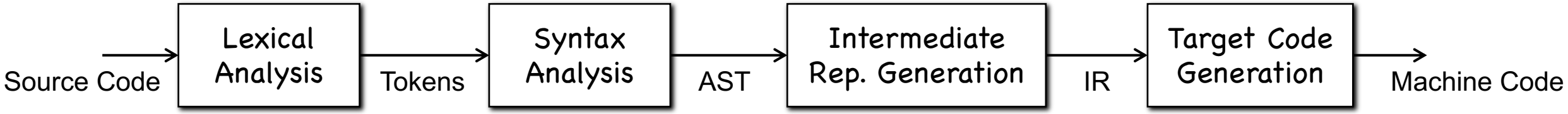
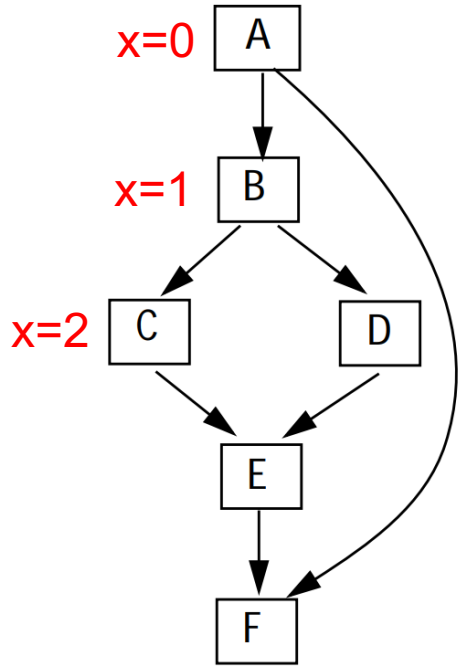


Block	A	B	C	D	E	F
Dominance Frontier	{ }	{F}	{E}	{E}	{F}	{ }



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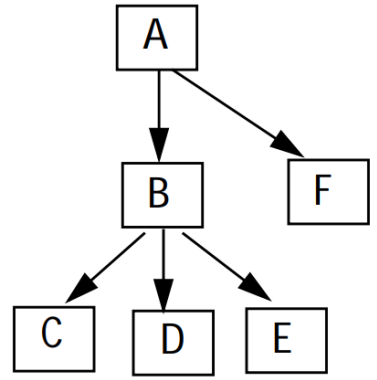

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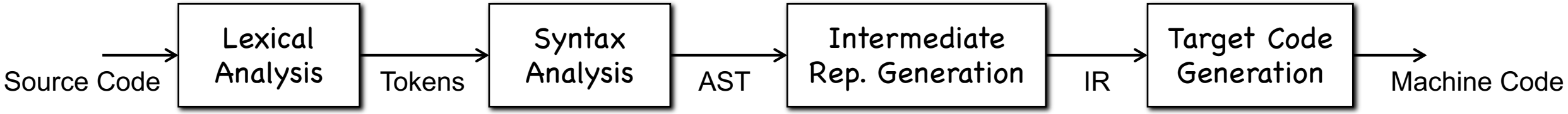
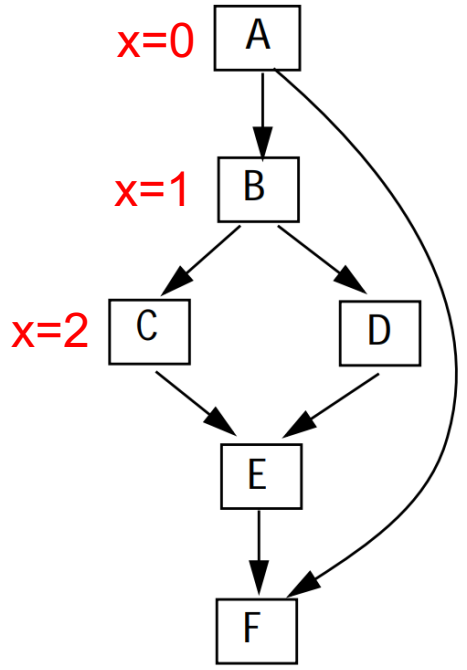


Block	A	B	C	D	E	F
Dominance Frontier	{ }	{F}	{E}	{E}	{F}	{ }

IDF({ A, B, C }) = { E, F }



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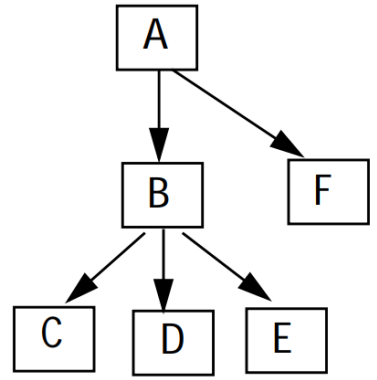

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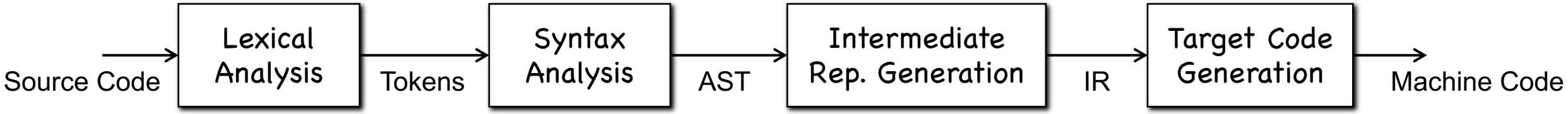
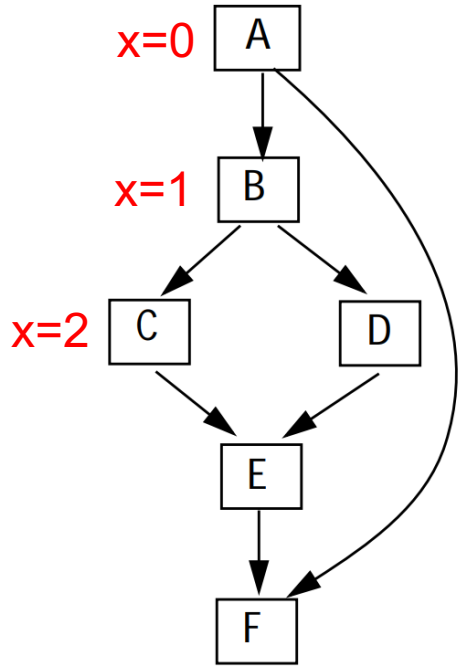


Block	A	B	C	D	E	F
Dominance Frontier	{ }	{F}	{E}	{E}	{F}	{ }

IDF({ A, B, C }) = { E, F }

Step 1: Insert **ϕ** at all blocks in IDF



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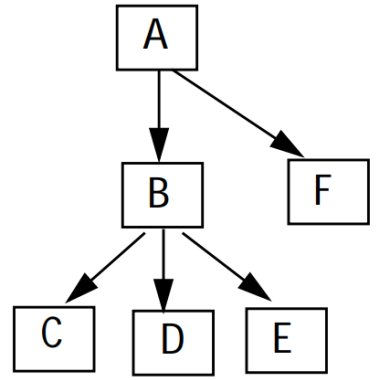

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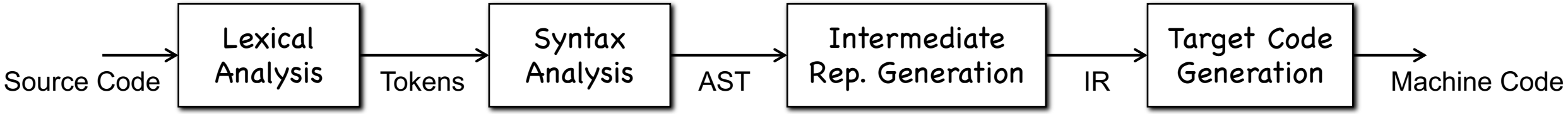

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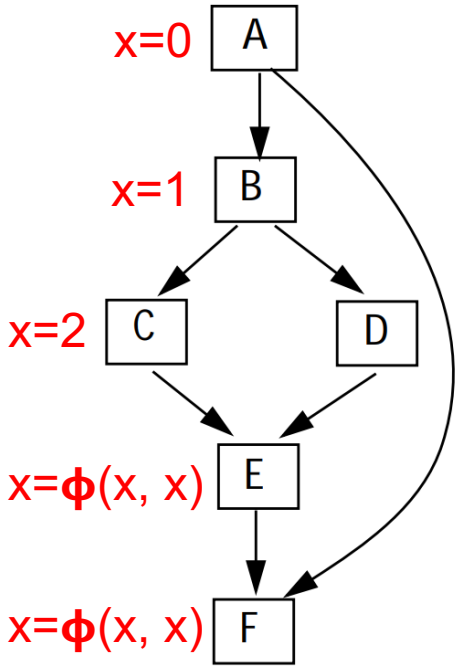

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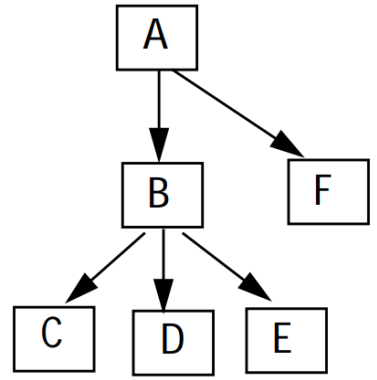

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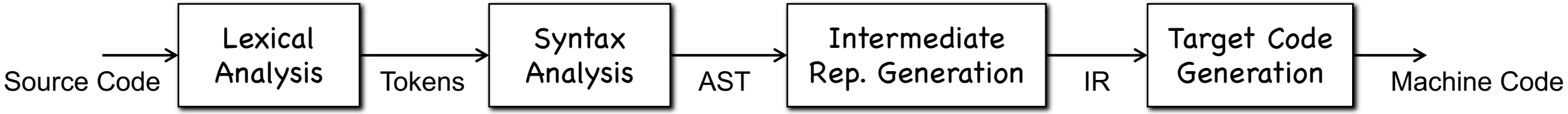

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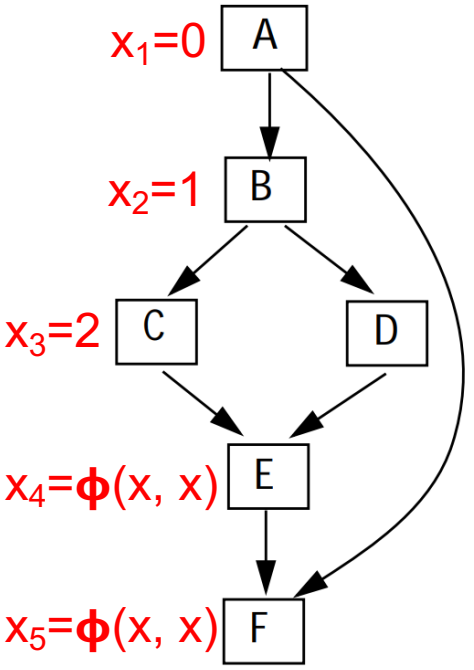

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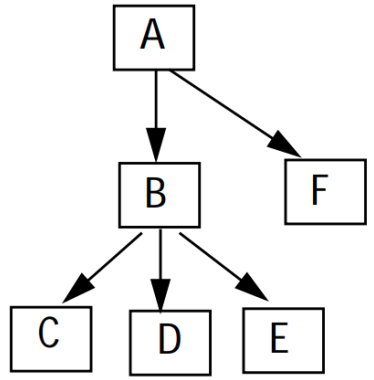

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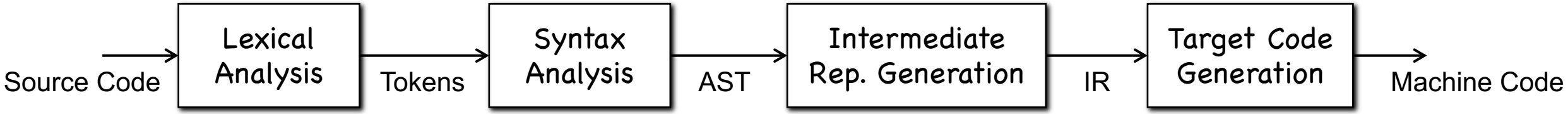

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Step 1: Insert **ϕ** at all blocks in IDF
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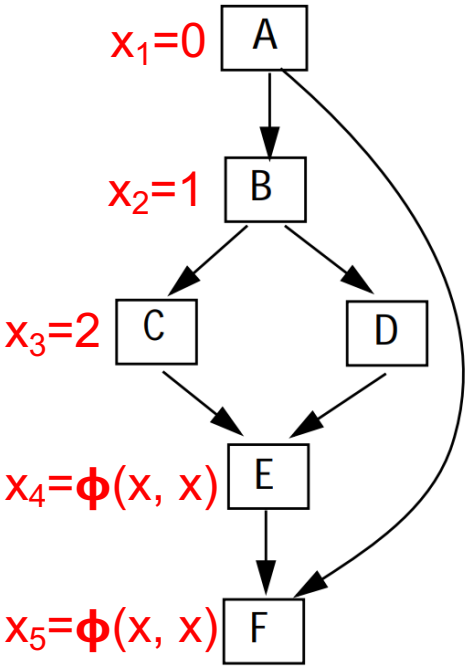
For each use of **x**, find proper def by climbing DT



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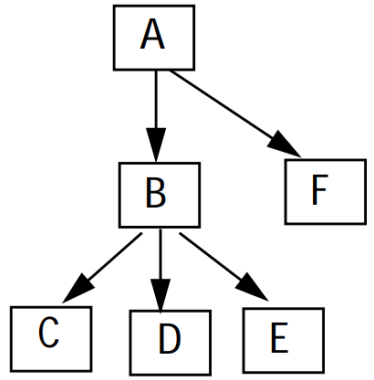

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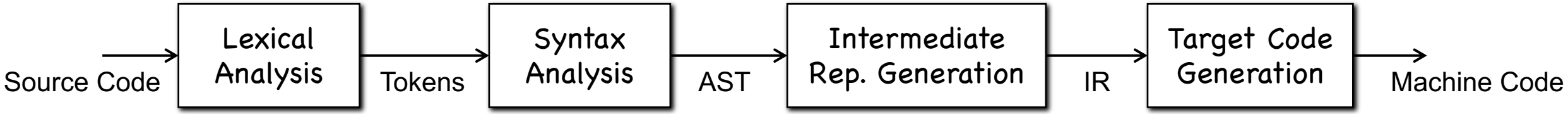

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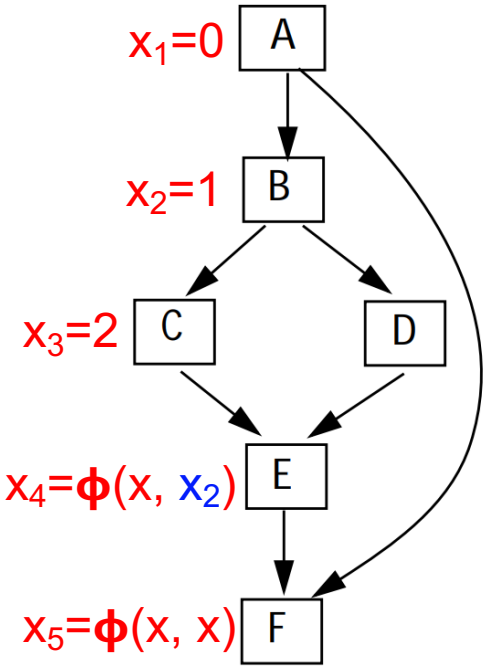

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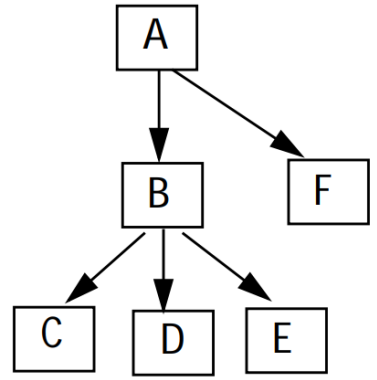

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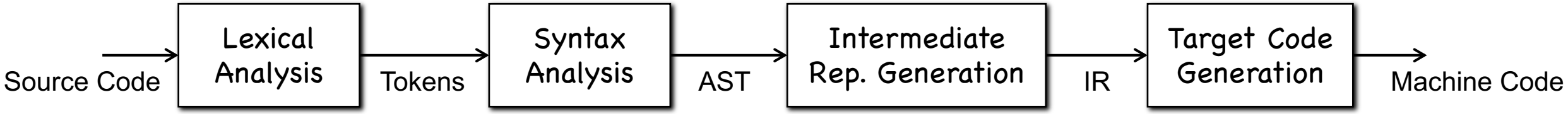

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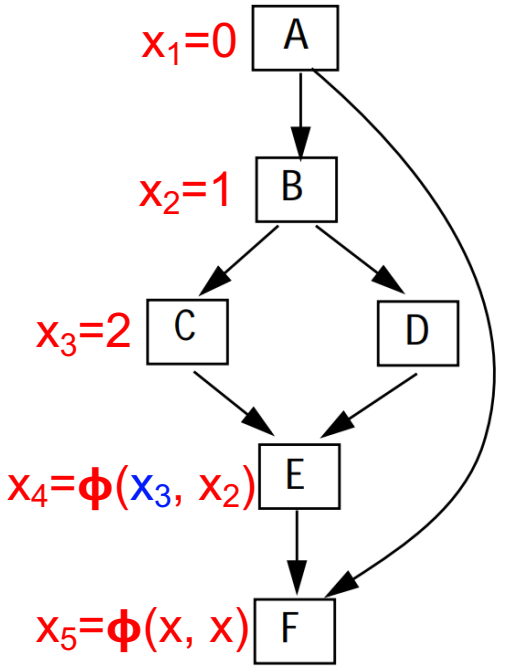

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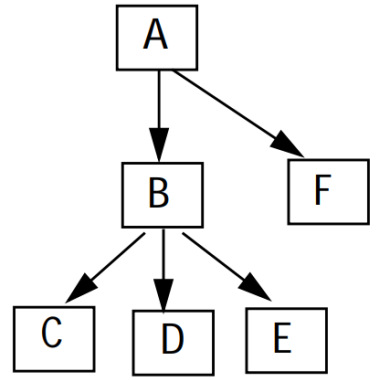

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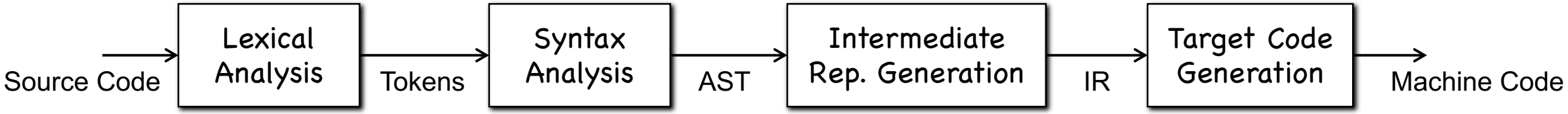

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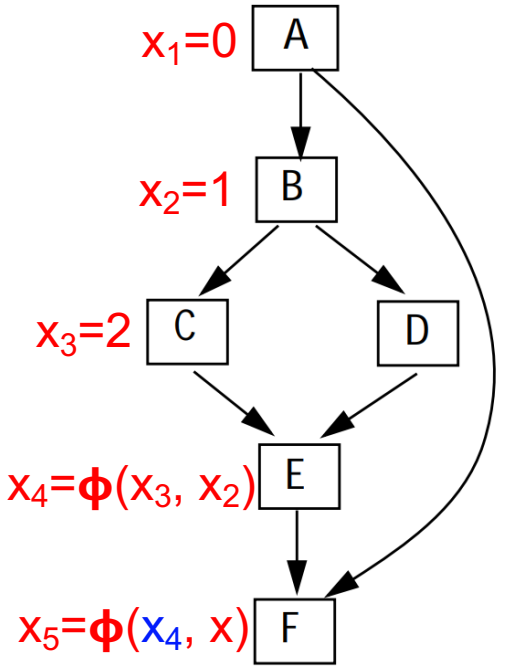

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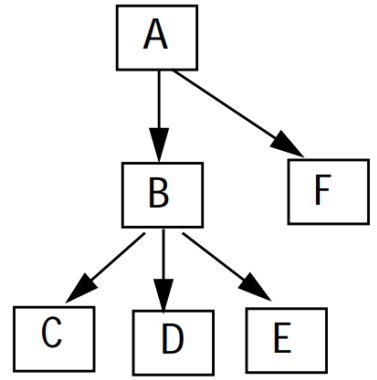

Dominator Tree

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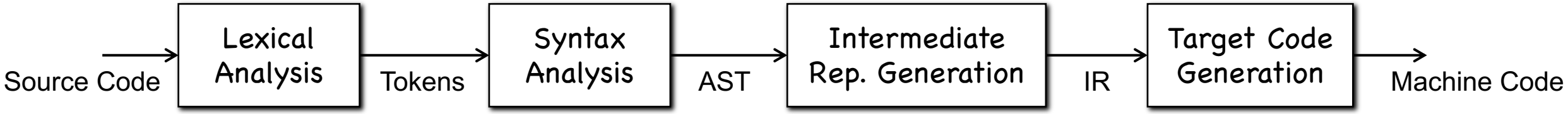

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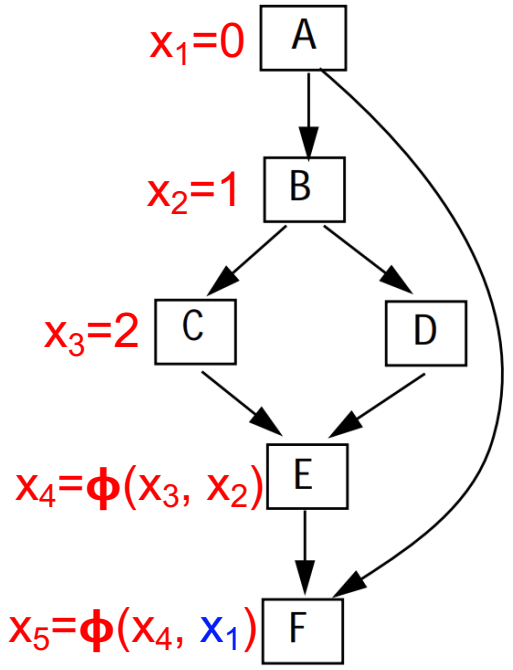

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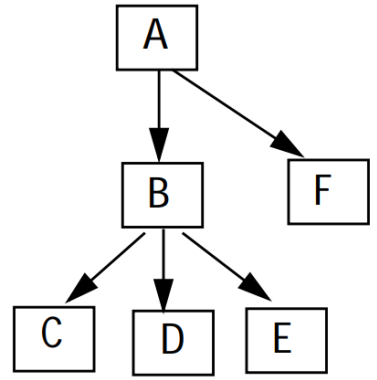
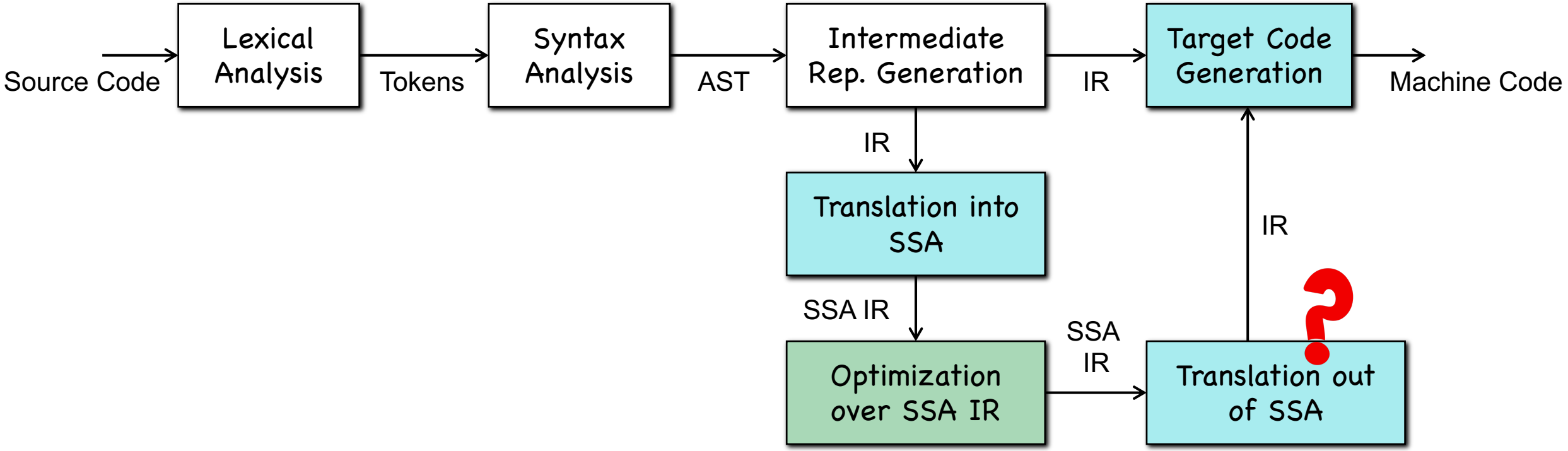
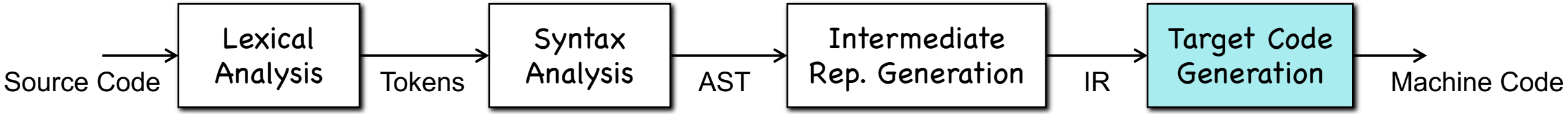


Dominator Tree

Recap: SSA in Compilers



Target Code Generation



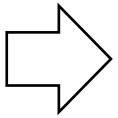

```
b = a + 5

c = b * 1

c = 2 + c

d = c
```

Three-Address Code



```
LD   R0,  a
ADD  R1,  R0,  5

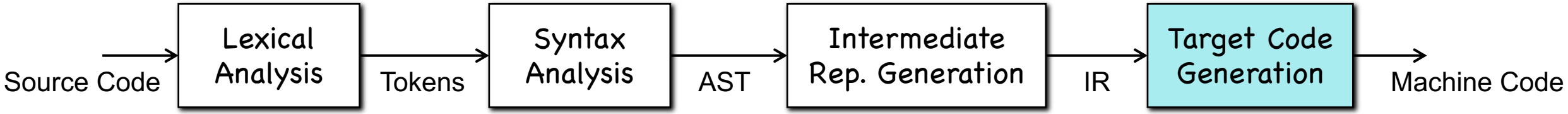
ST   b,   R1
ST   c,   R1

ADD  R2,  R1,  2
ST   c,   R2

LD   R3,  c
ST   d,   R3
```

Machine Code

Target Code Generation



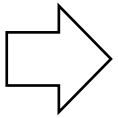

```
b = a + 5

c1 = b * 1

c2 = 2 + c1

d = c2
```

Three-Address Code (SSA)



```
LD   R0,  a
ADD  R1,  R0,  5

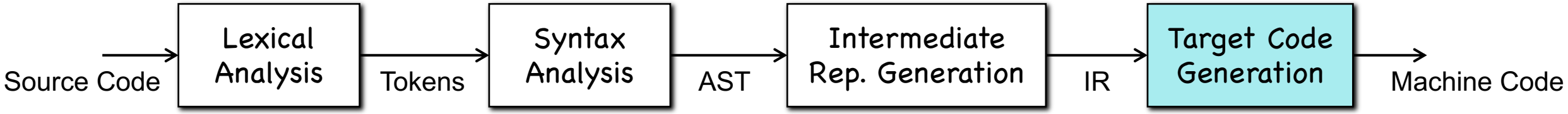
ST   b,   R1
ST   c1,  R1

ADD  R2,  R1,  2
ST   c2,  R2

LD   R3,  c2
ST   d,   R3
```

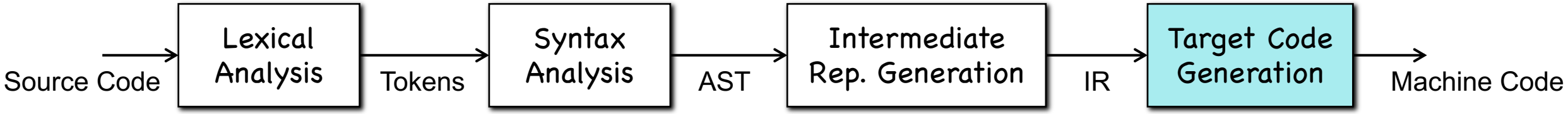
Machine Code

Target Code Generation



What challenges do we have to address, so as to translate SSA IR (e.g., LLVM IR) to machine code?

Target Code Generation



What challenges do we have to address, so as to translate SSA IR (e.g., LLVM IR) to machine code?

The ϕ functions!

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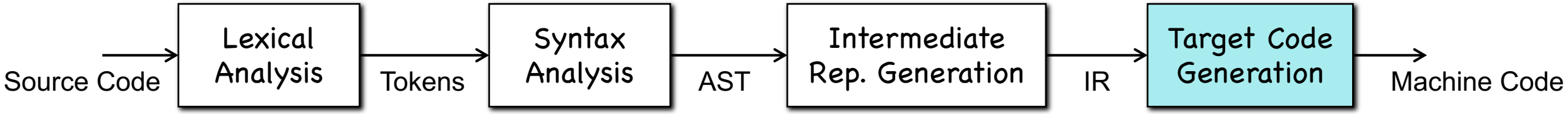
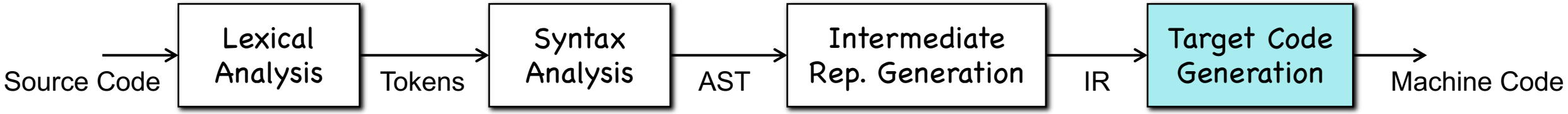


What challenges do we have to address, so as to translate SSA IR (e.g., LLVM IR) to machine code?

The ϕ functions!

How can we translate (or eliminate) the ϕ functions?

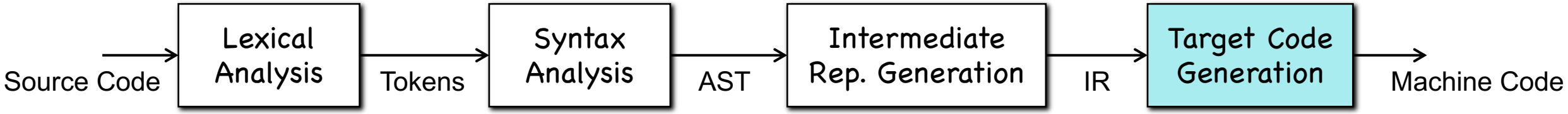
Target Code Generation




```
if ( flag ) x1 = -1; else x2 = 1;
x3 = ϕ (x1, x2);
y = x3 * a
```

Try!

Target Code Generation



Solution 1: Since ϕ merge values of the same variable, let's replace variables merged by ϕ with the same variable.

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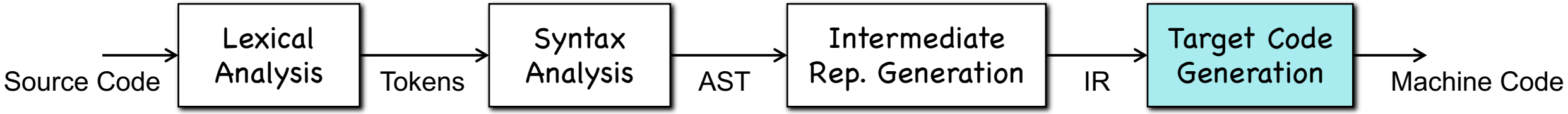

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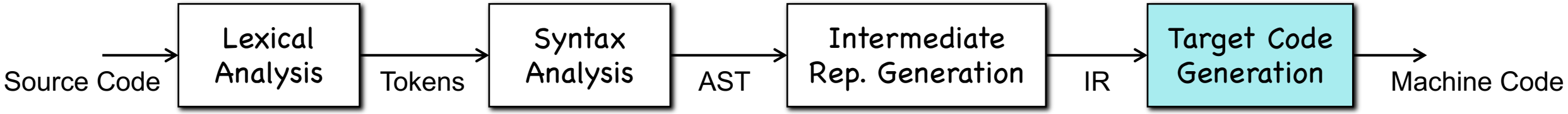

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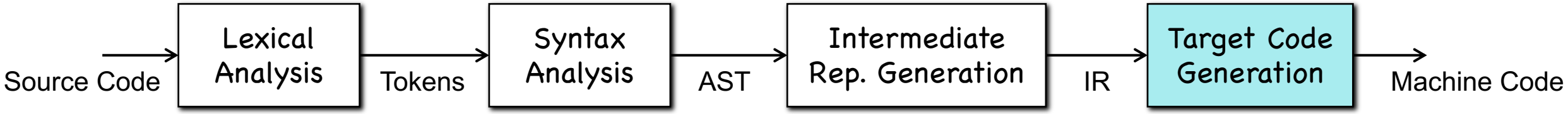

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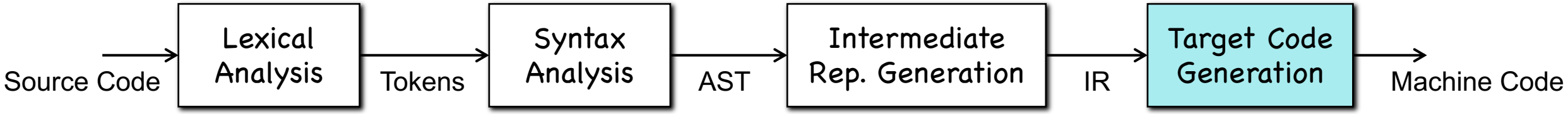

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```
x0 = 1;
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    x1 = ϕ(x0, x2);
    x2 = x1 + 1;
} while(…);
print(x1);
```

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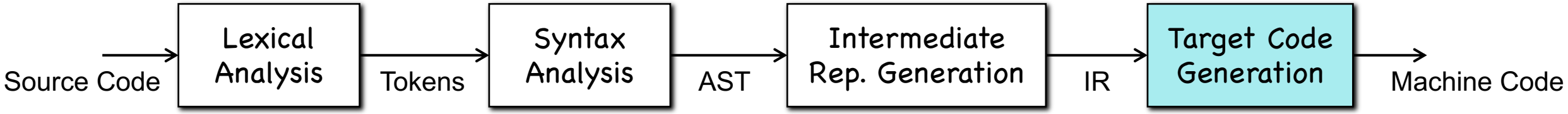

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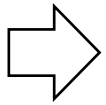

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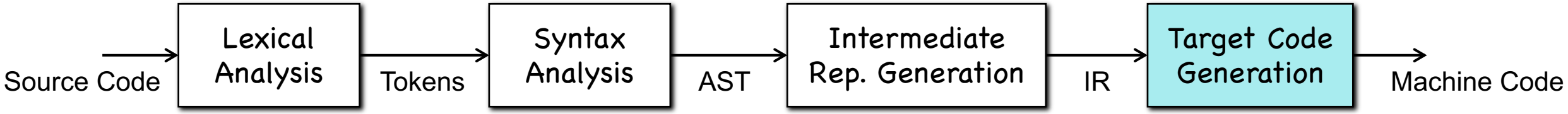
x = 1;
do {
    x1 = ϕ(x0, x2);
    x = x + 1;
} while(…);
print(x);
```

Target Code Generation



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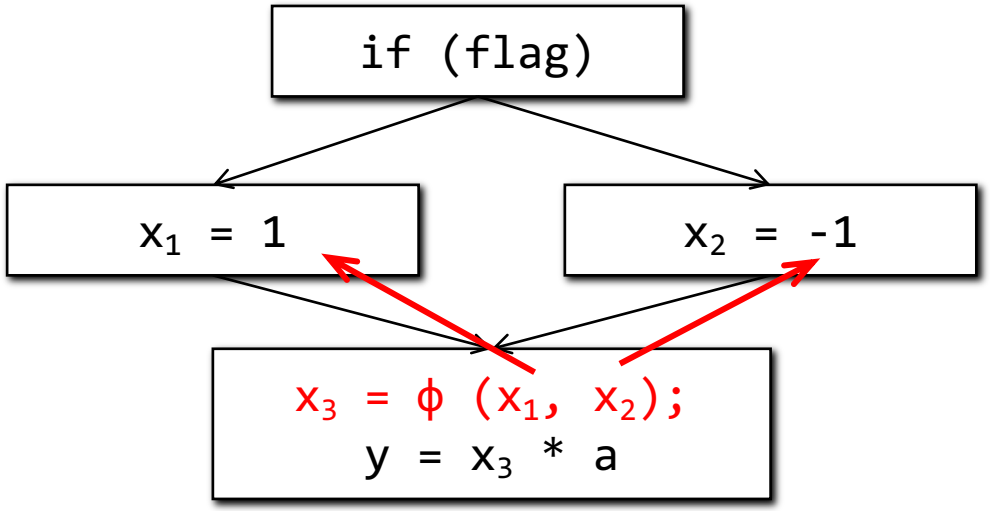

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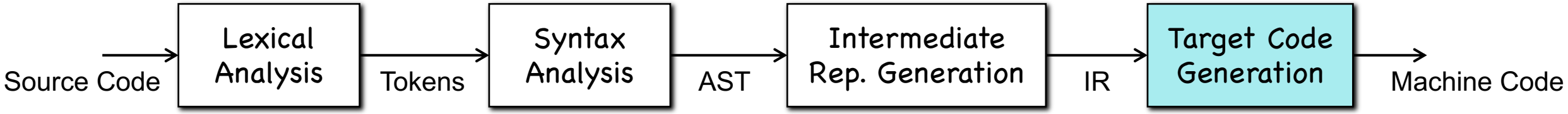

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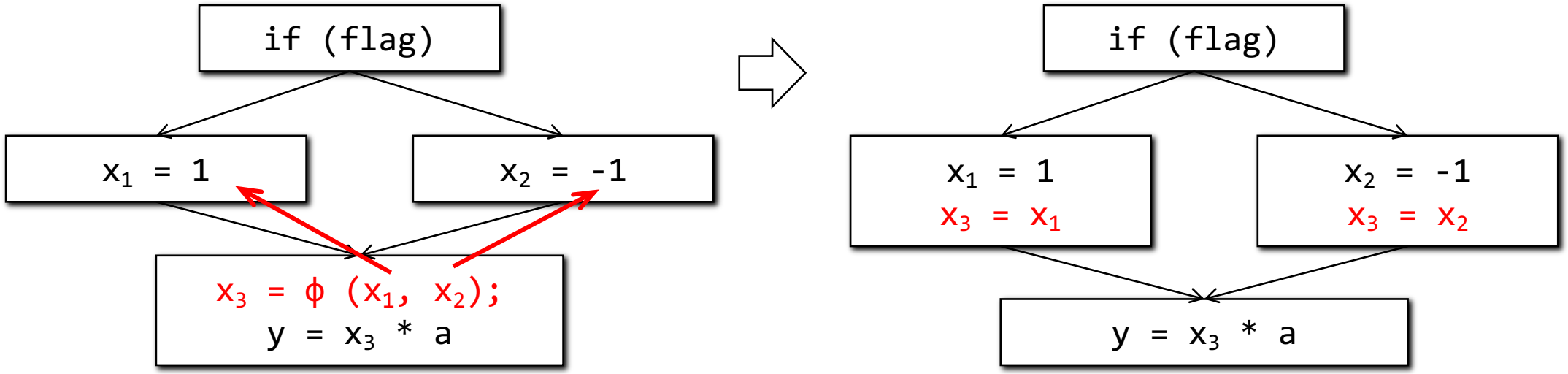

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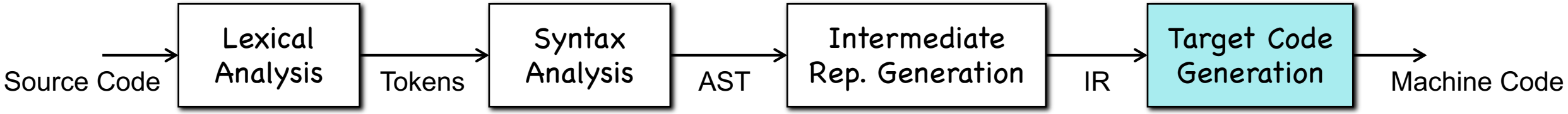

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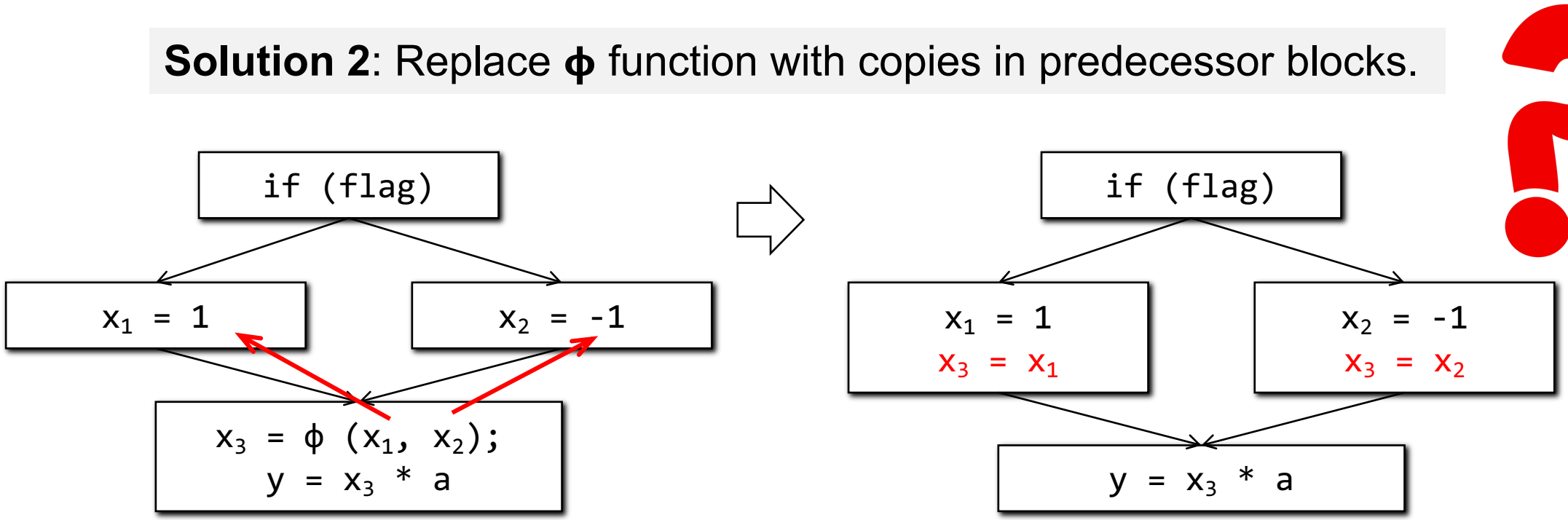

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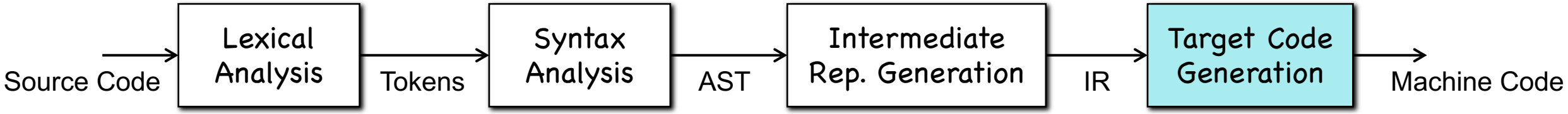

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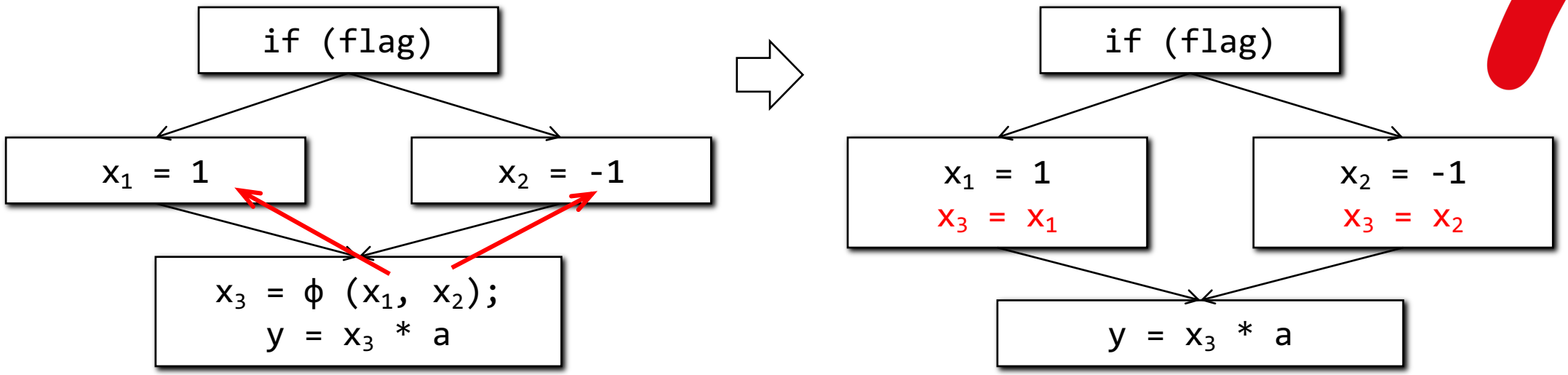

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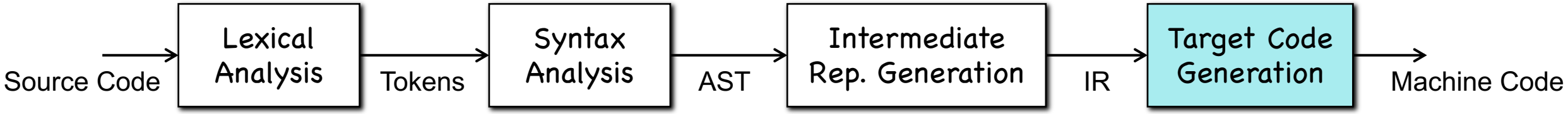

Solution 2: Replace **ϕ** function with copies in predecessor blocks.

```
x0 = 1;
do {
    x1 = ϕ(x0, x2);
    x2 = x1 + 1;

} while (…);
print(x1);
```

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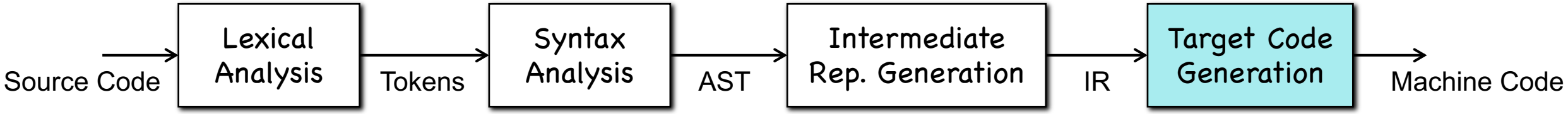

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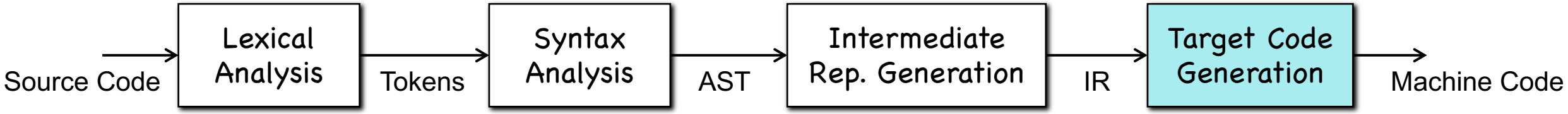

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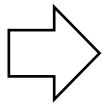

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x0 = 1; x1 = x0;
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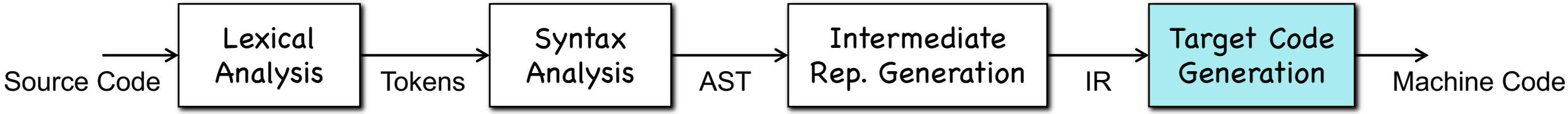

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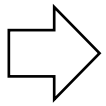

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```

The Lost Copy Problem!

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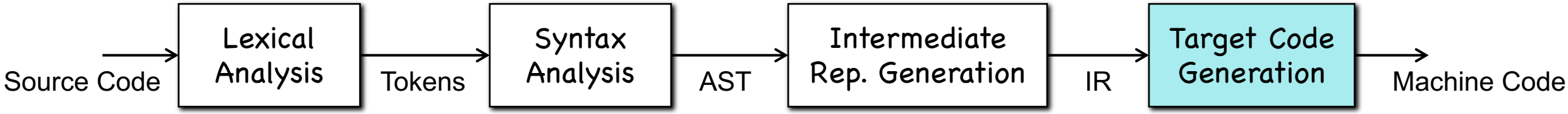

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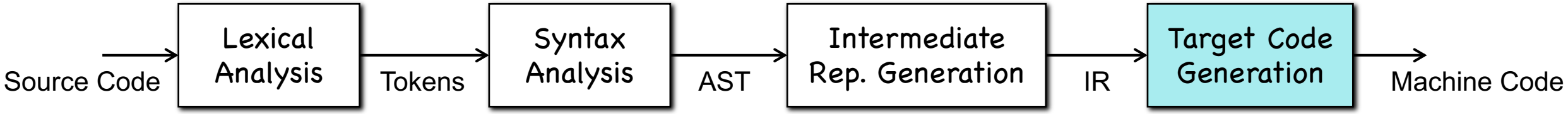
x0 = 1; y0 = 2;

do {
    x1 = ϕ(x0, y1);
    y1 = ϕ(y0, x1);
} while (…);
print(x1, y1);
```

Target Code Generation



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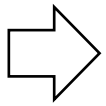

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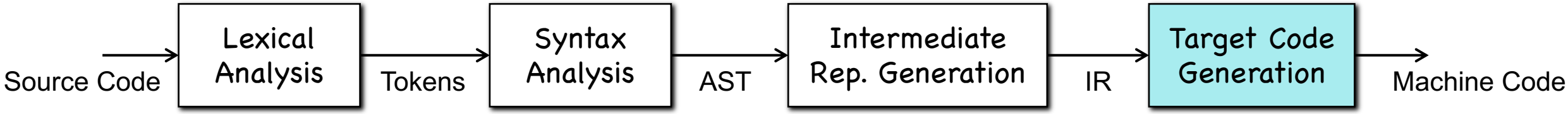

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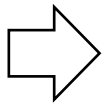

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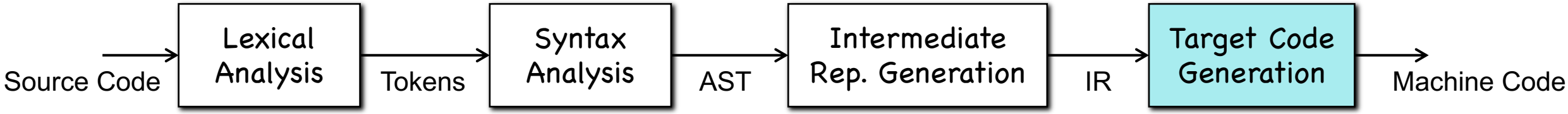
```

The Swap Problem!

Target Code Generation



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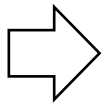

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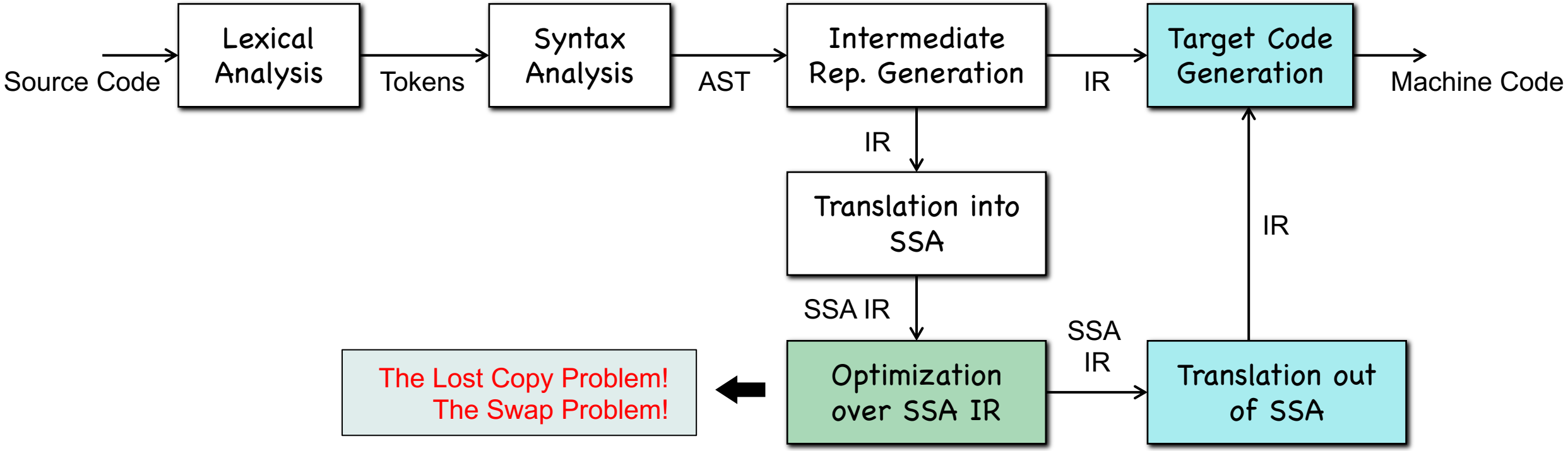
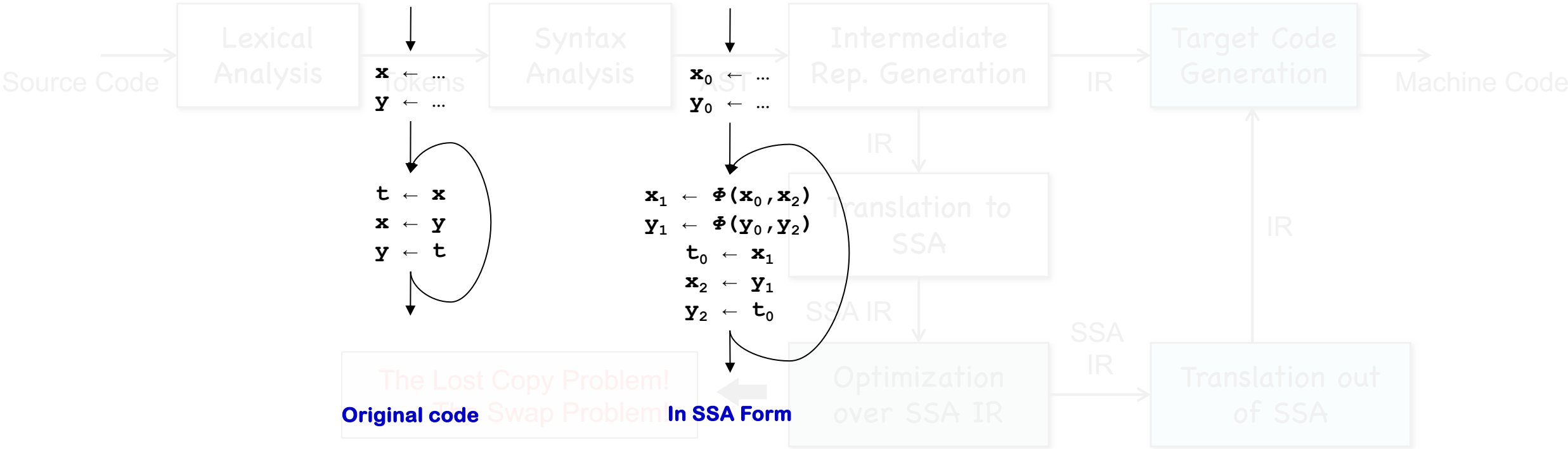
The Swap Problem!

Where're the Problems from?



Where're the Problems from?



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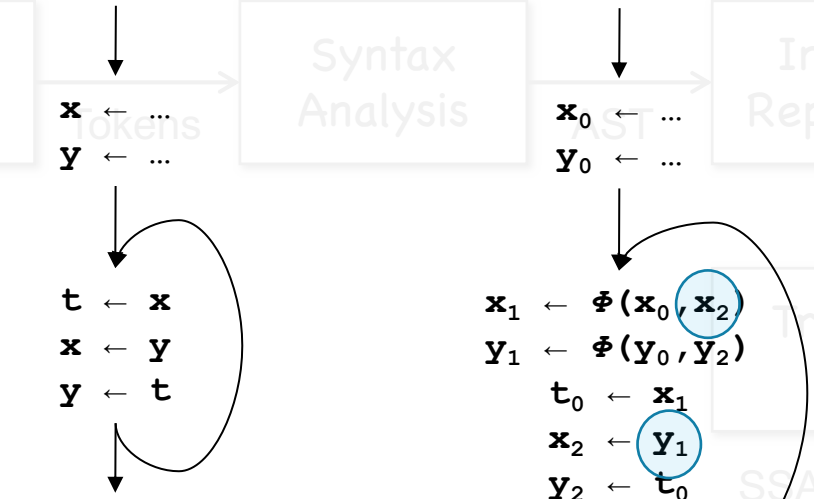
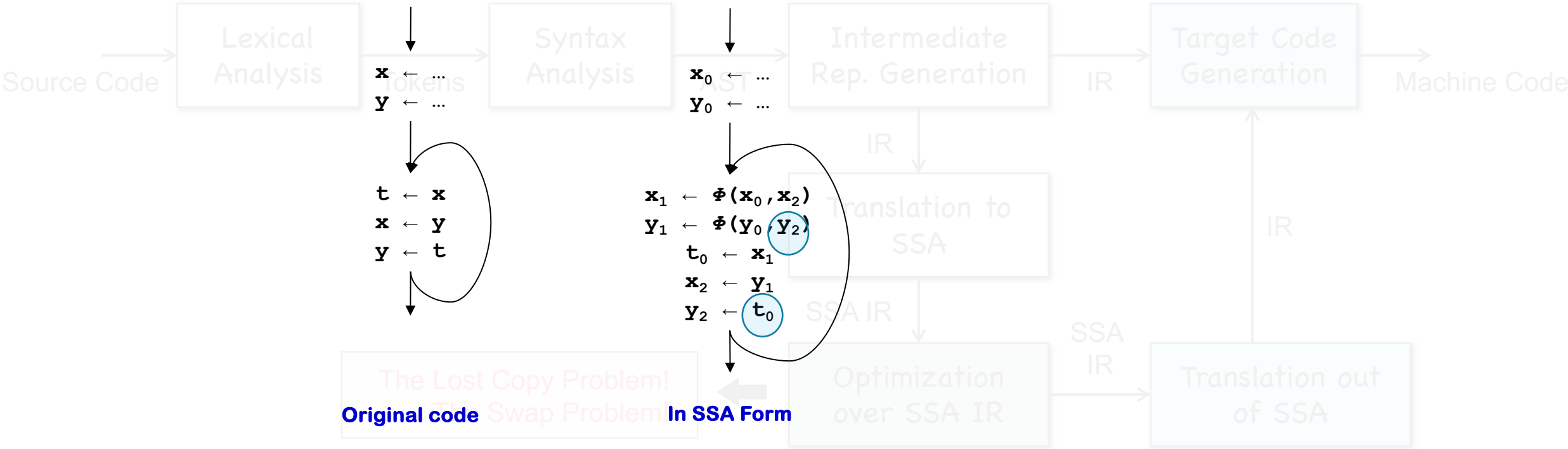


Original code In SSA Form

Where're the Problems from?



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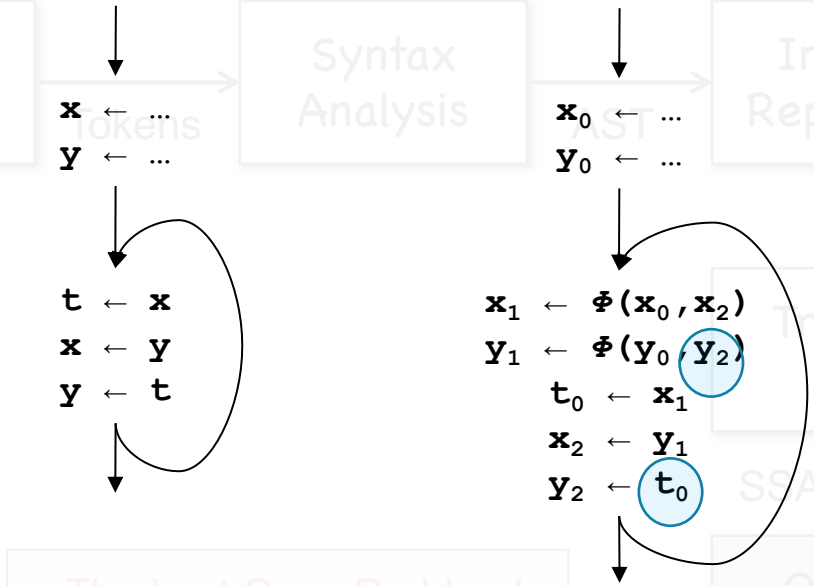
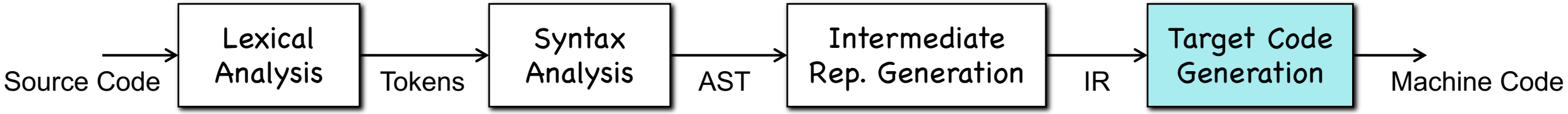


Target Code Generation



- These problems were identified by **Cliff Click** and were first published by **Briggs et. al.** in 1997. They address the issues by ad-hoc ways.

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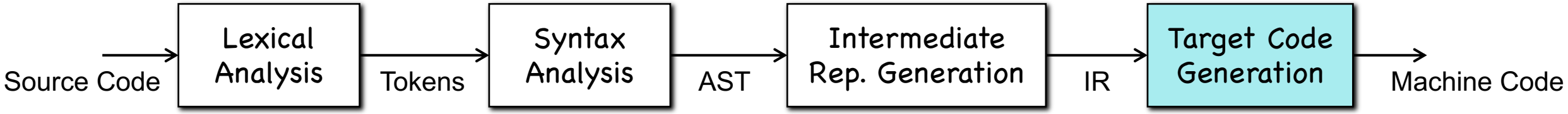

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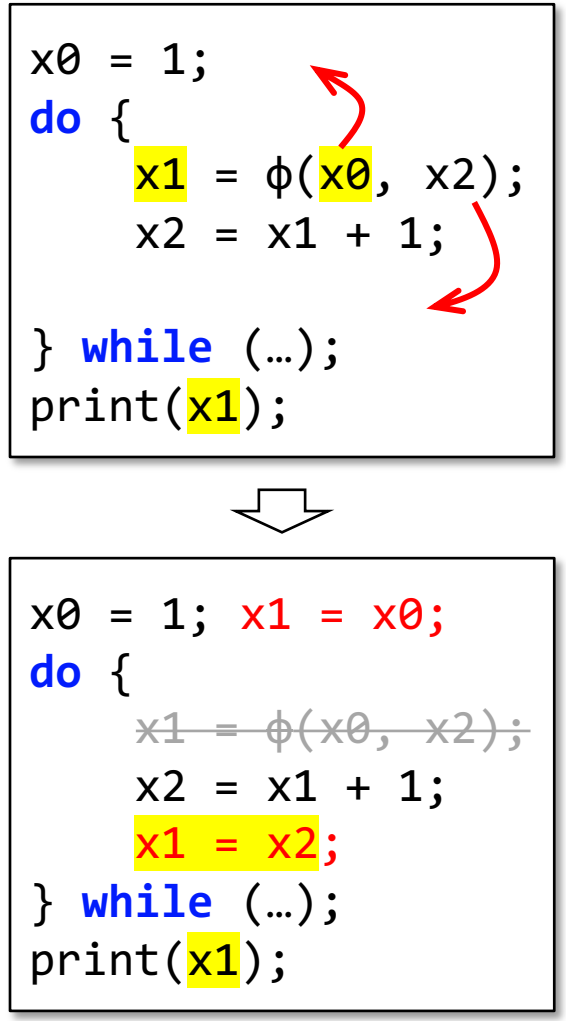
- Revisiting Out-of-SSA Translation for Correctness, Code Quality and Efficiency.
- Published in CGO 2009. By **Boissinot et. al.**

The Lost Copy Problem

- The copy X1 = X0 is lost!
- To address the issue, we need to create a temp variable to hold this value.



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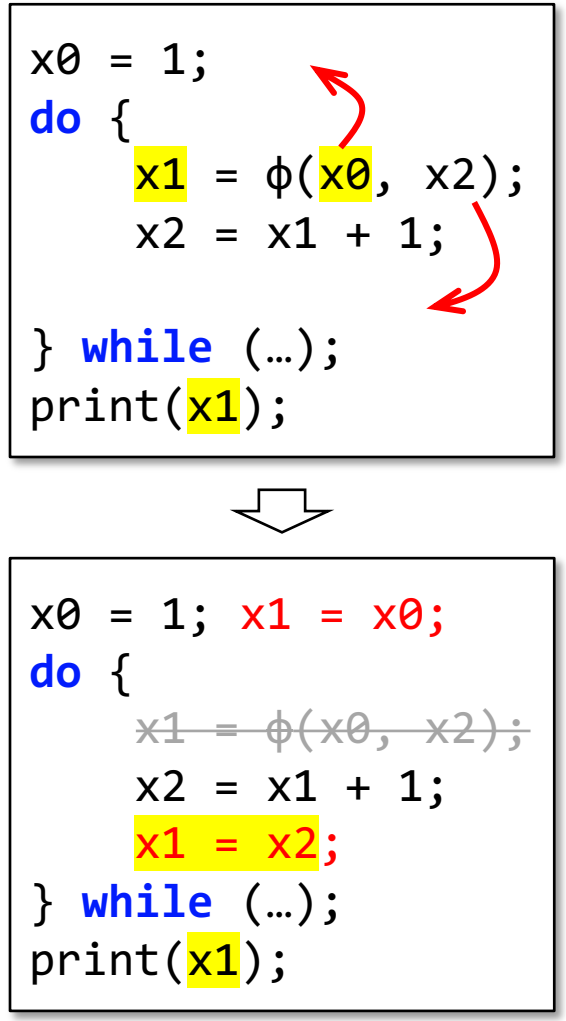

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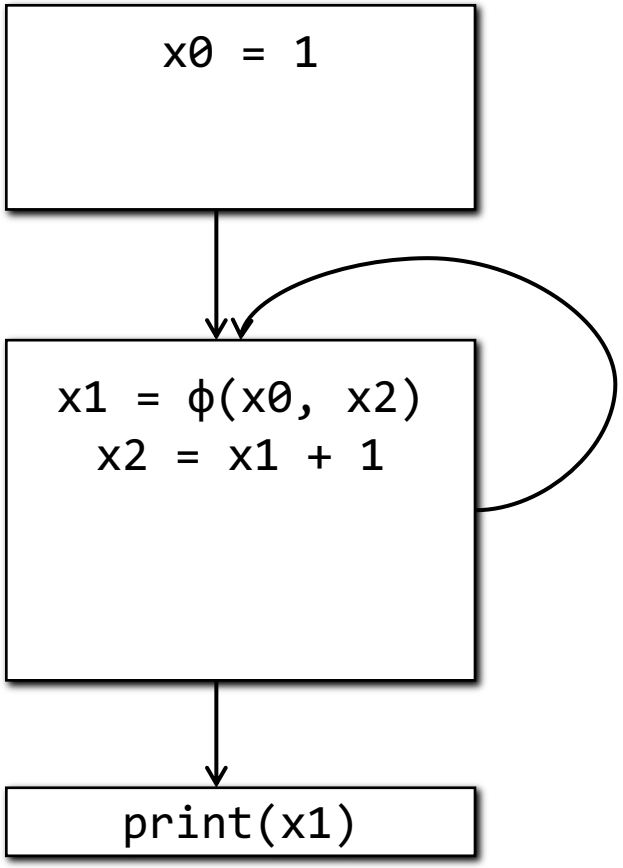

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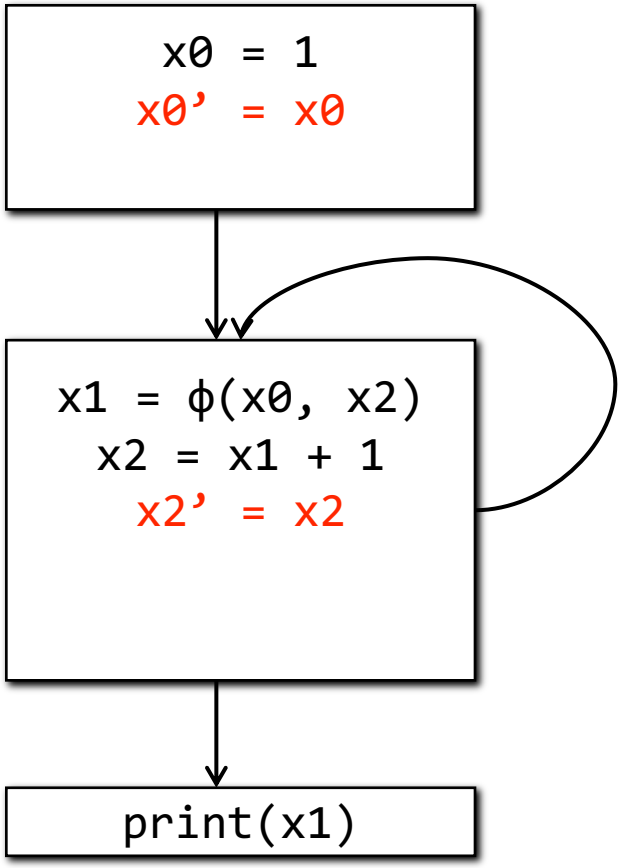

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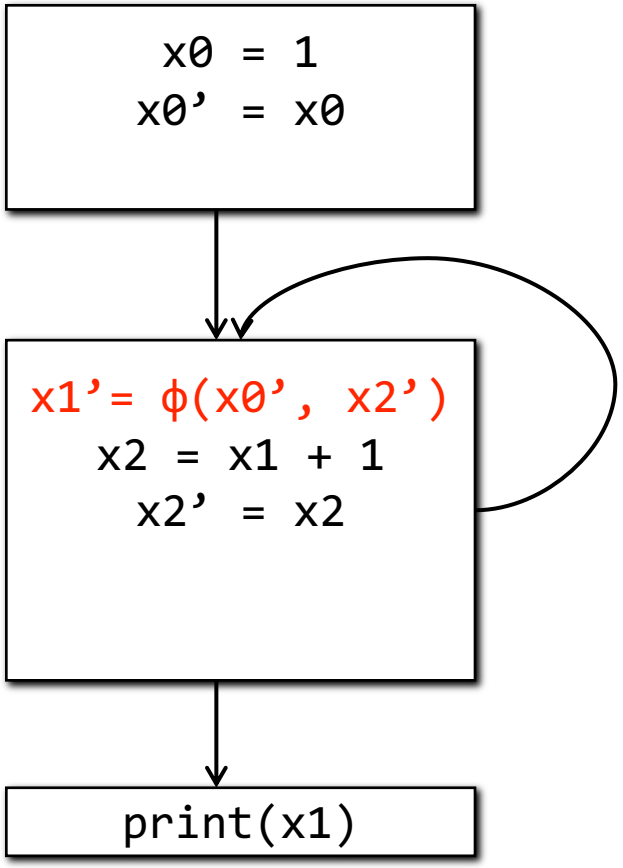

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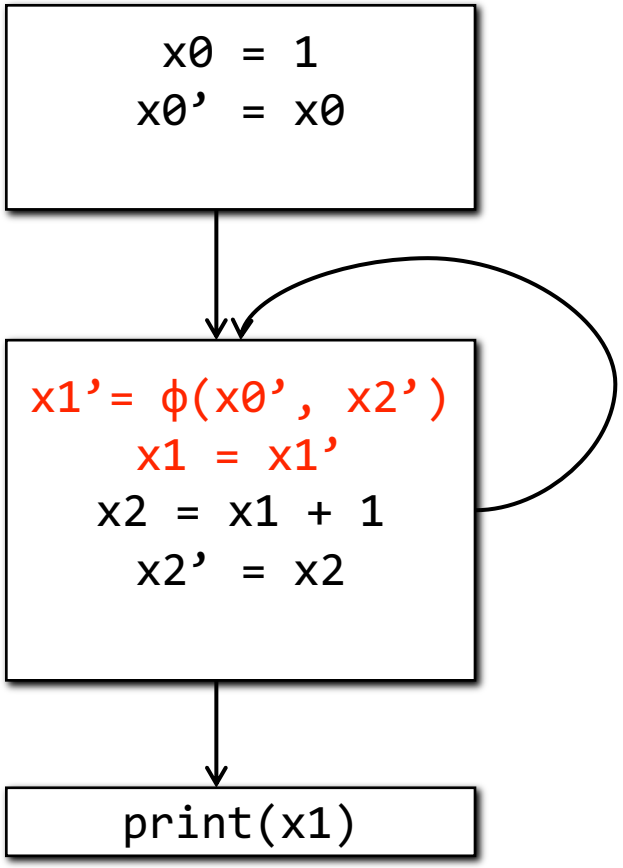

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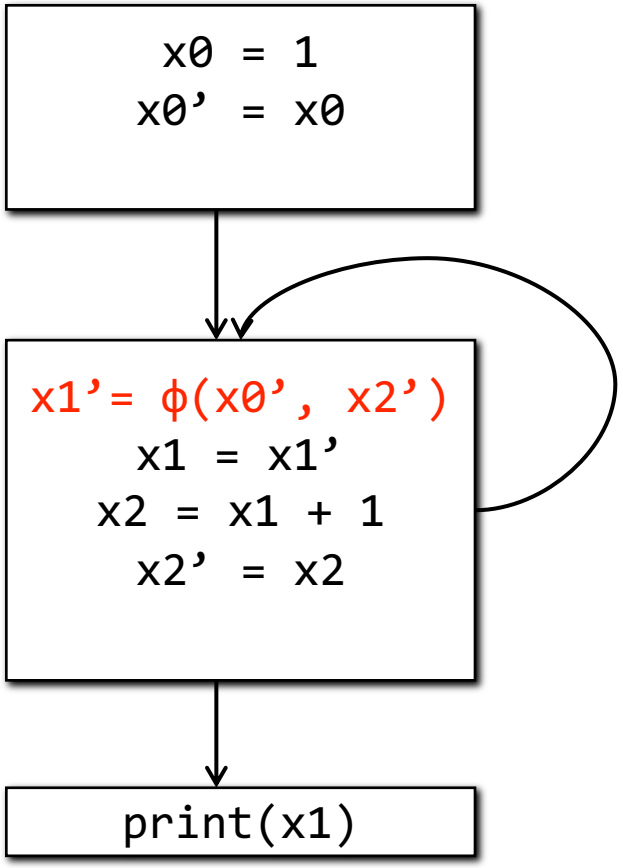

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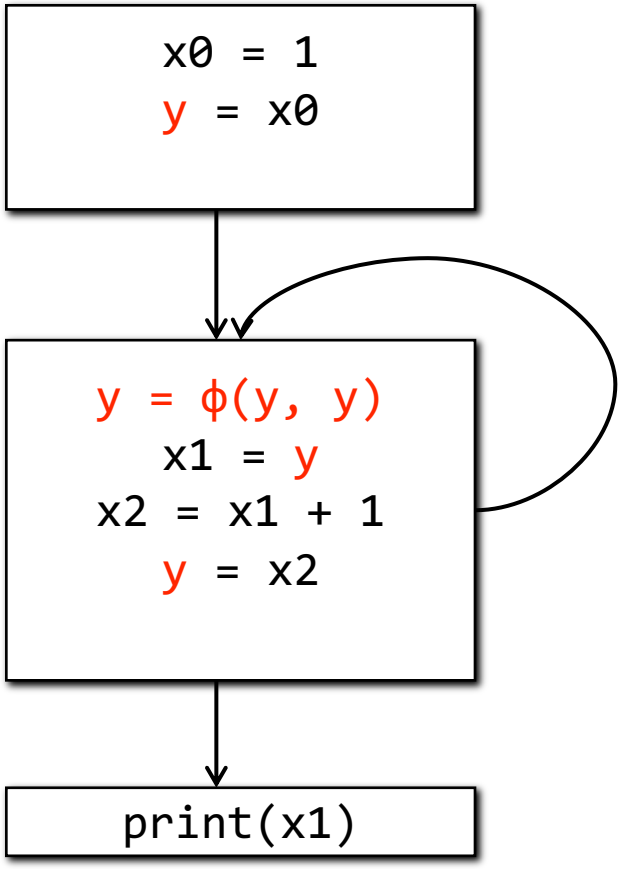

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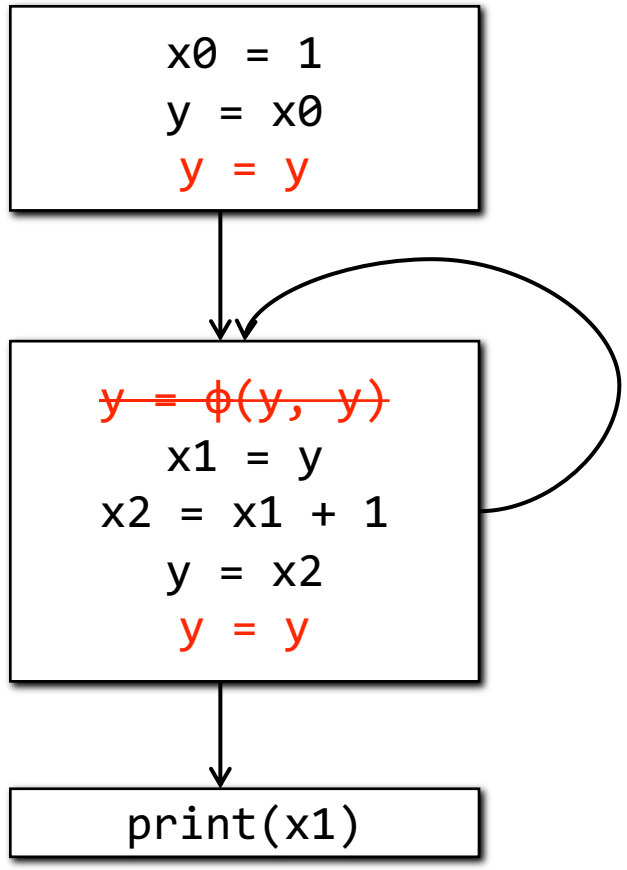

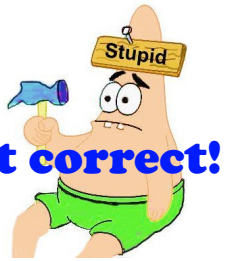

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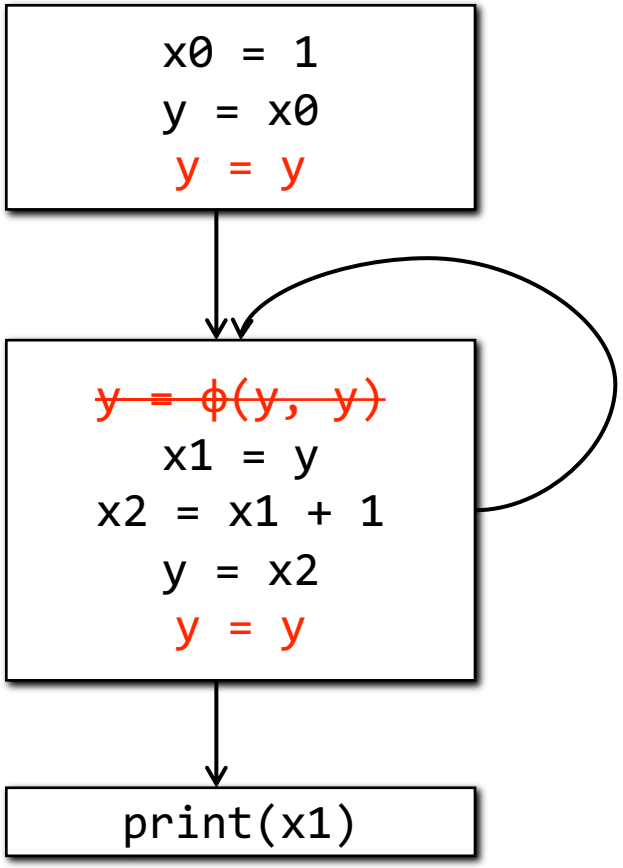

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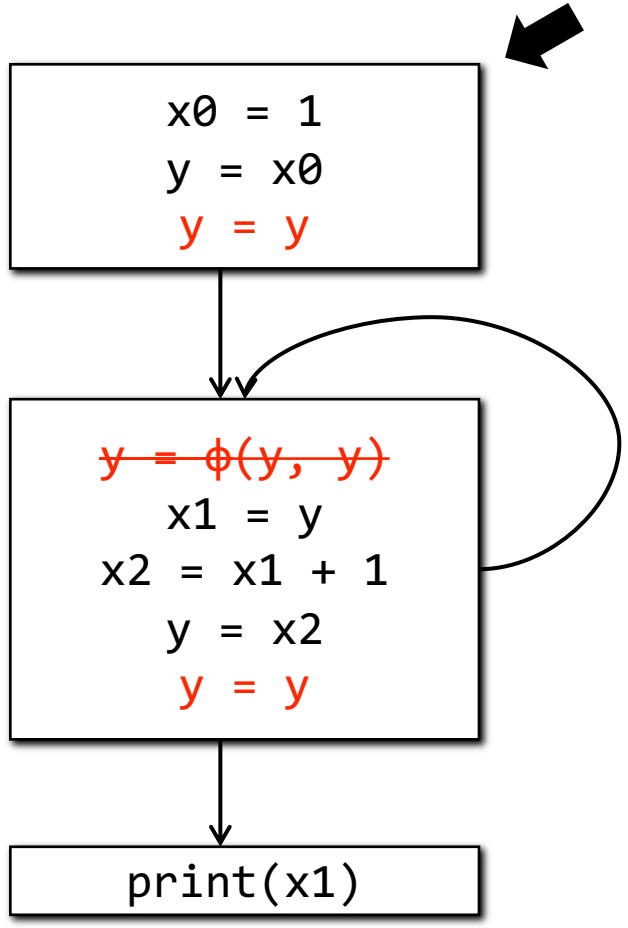

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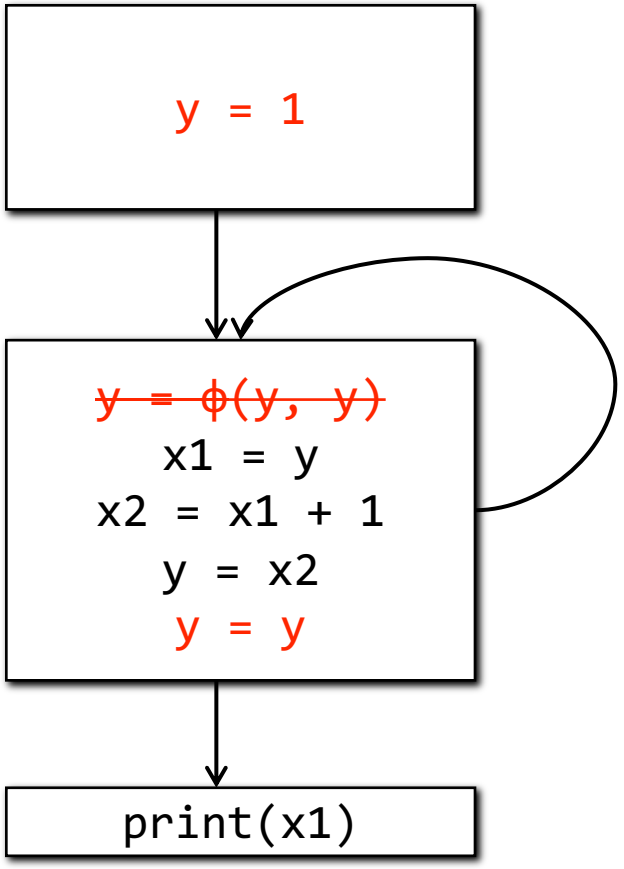

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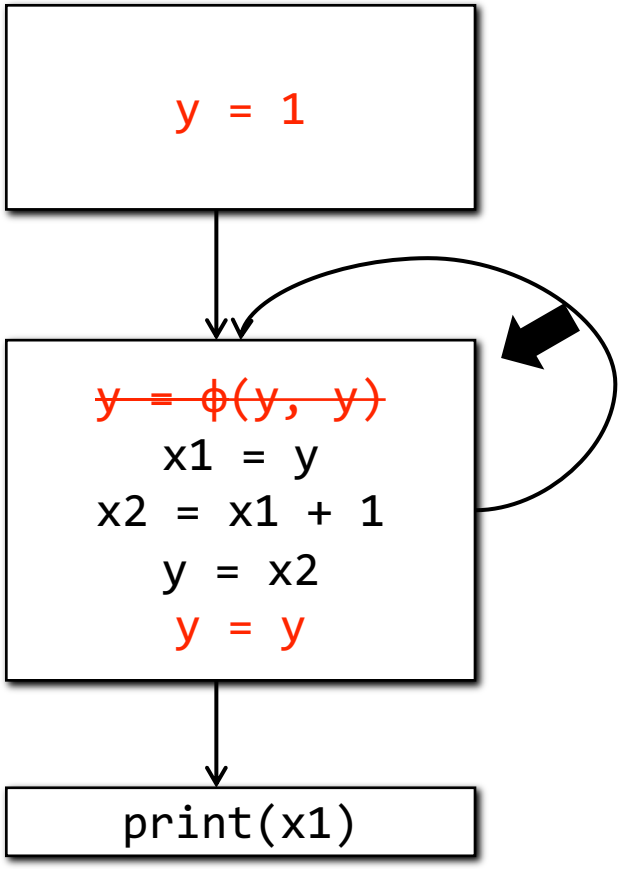

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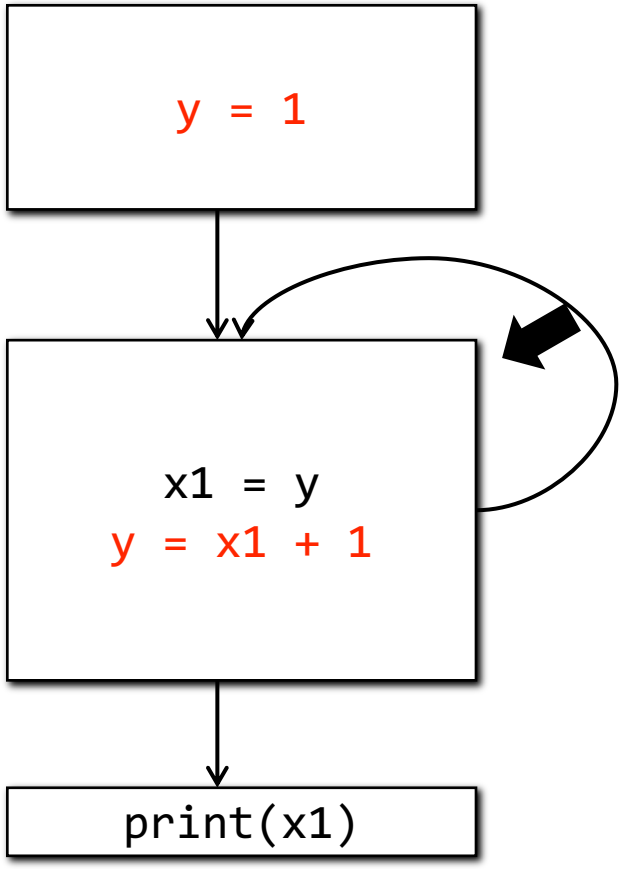

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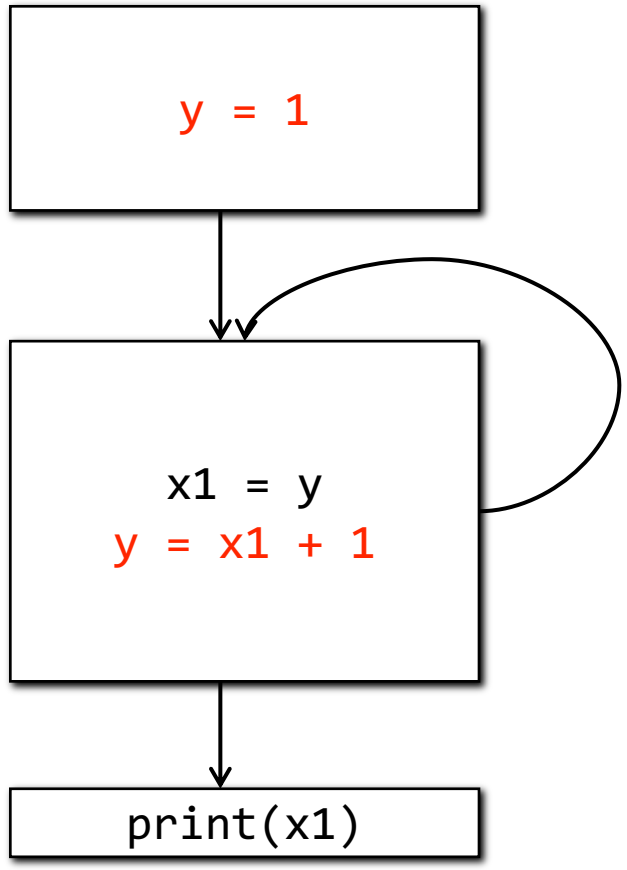
```
x0 = 1;
do {
    x1 = ϕ(x0, x2);
    x2 = x1 + 1;

} while (…);
print(x1);
```



The Swap Problem

- Caused by multiple ϕ in the same block
- The parallel semantics of ϕ statements

```
x0 = 1; y0 = 2;

do {
    x1 = ϕ(x0, y1);
    y1 = ϕ(y0, x1);
} while (…);
print(x1, y1);
```

```
x0 = 1; y0 = 2;
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do {
    x1 = y1;
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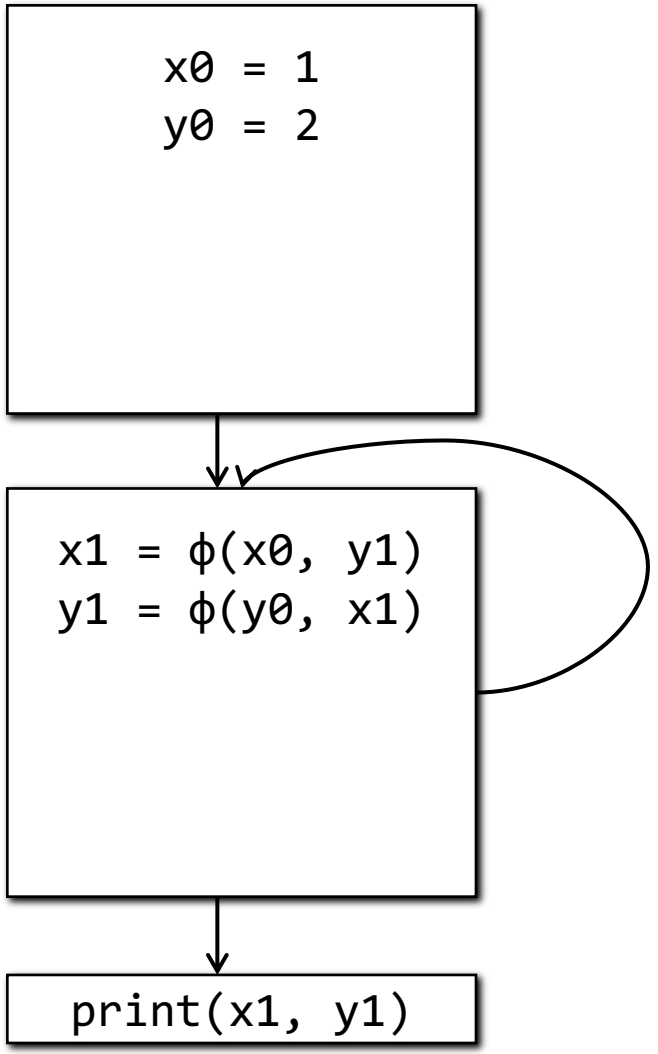

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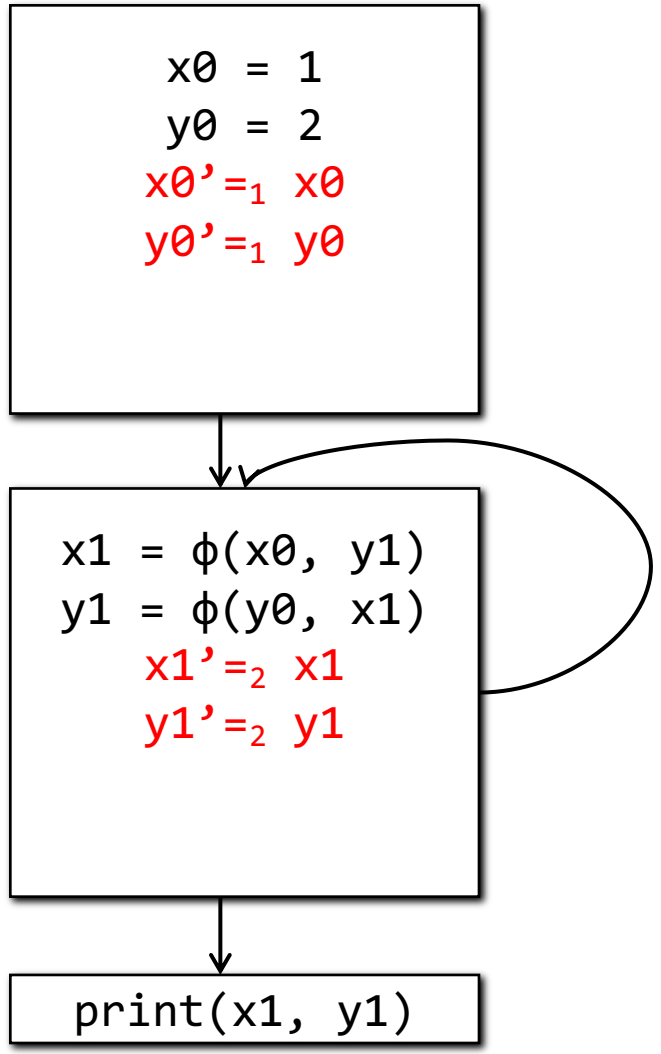

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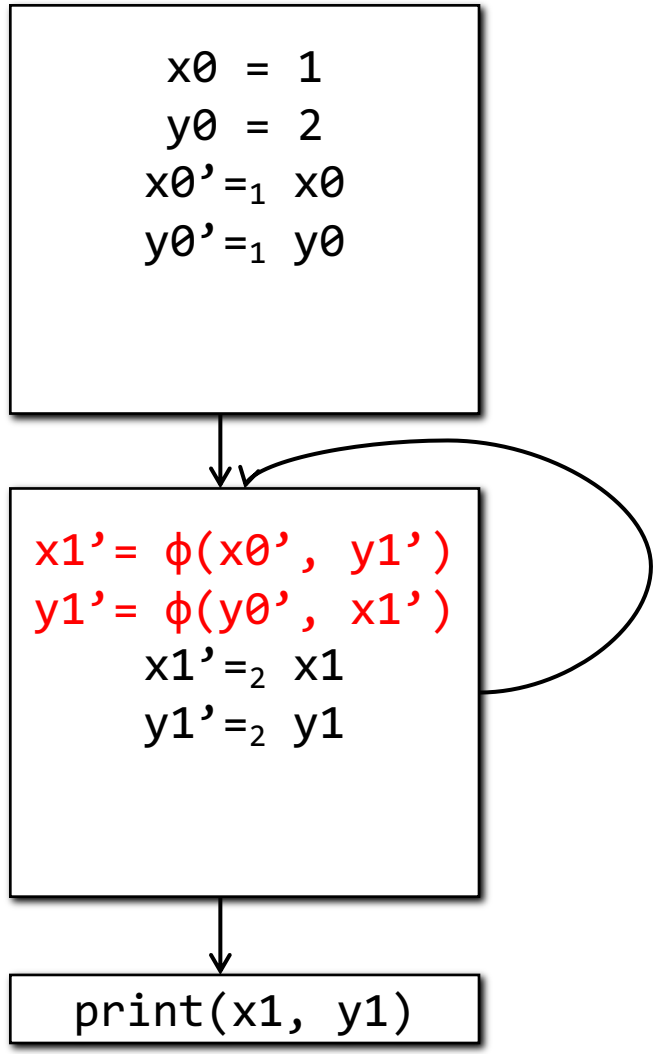

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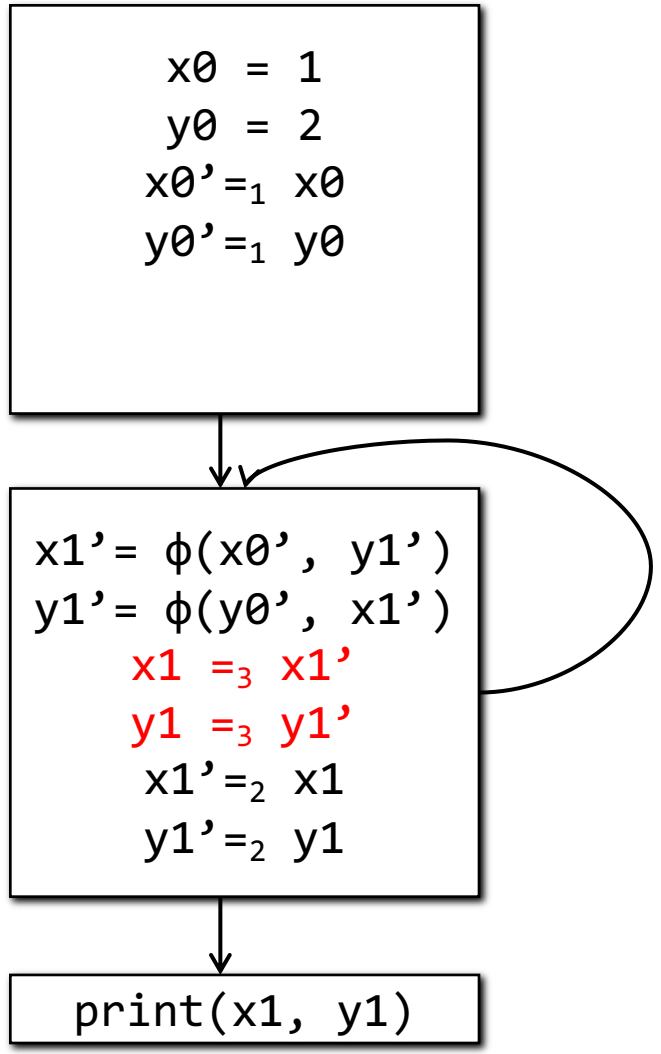

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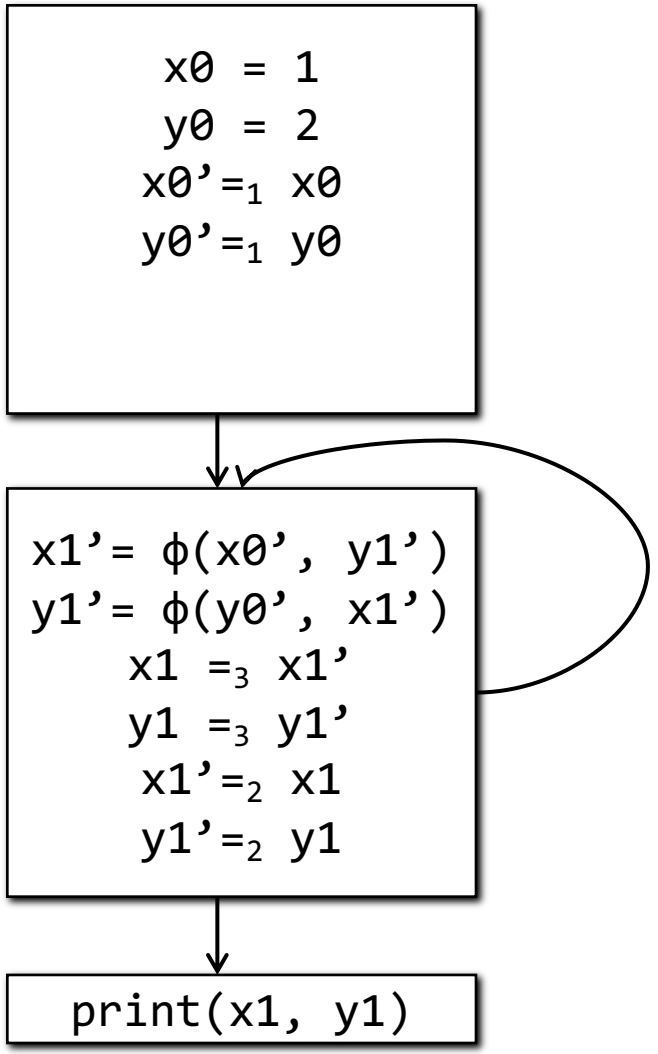

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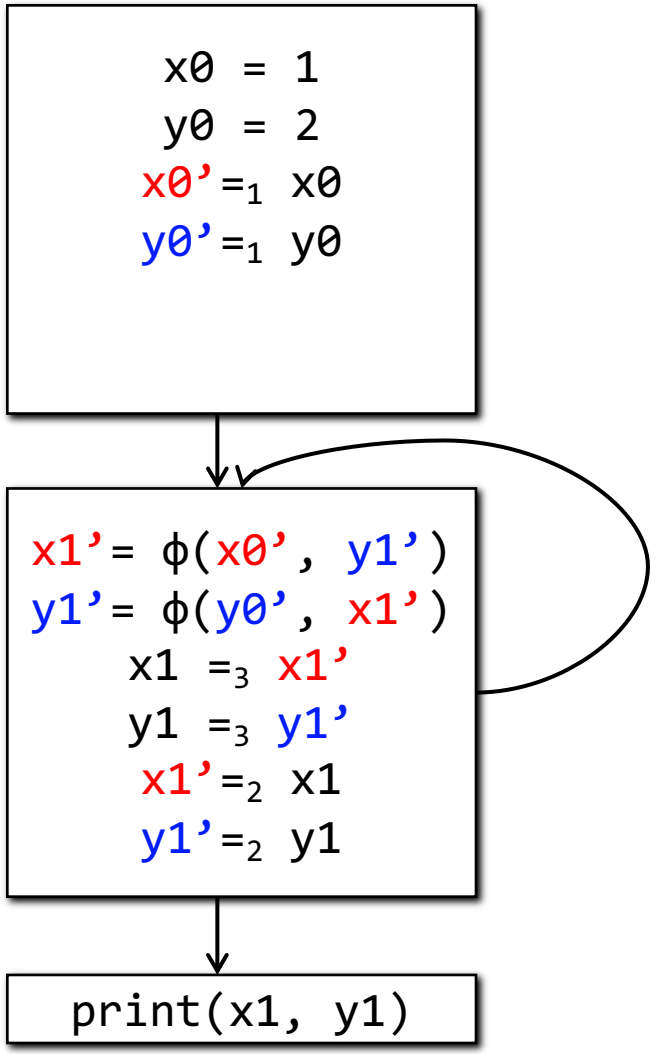

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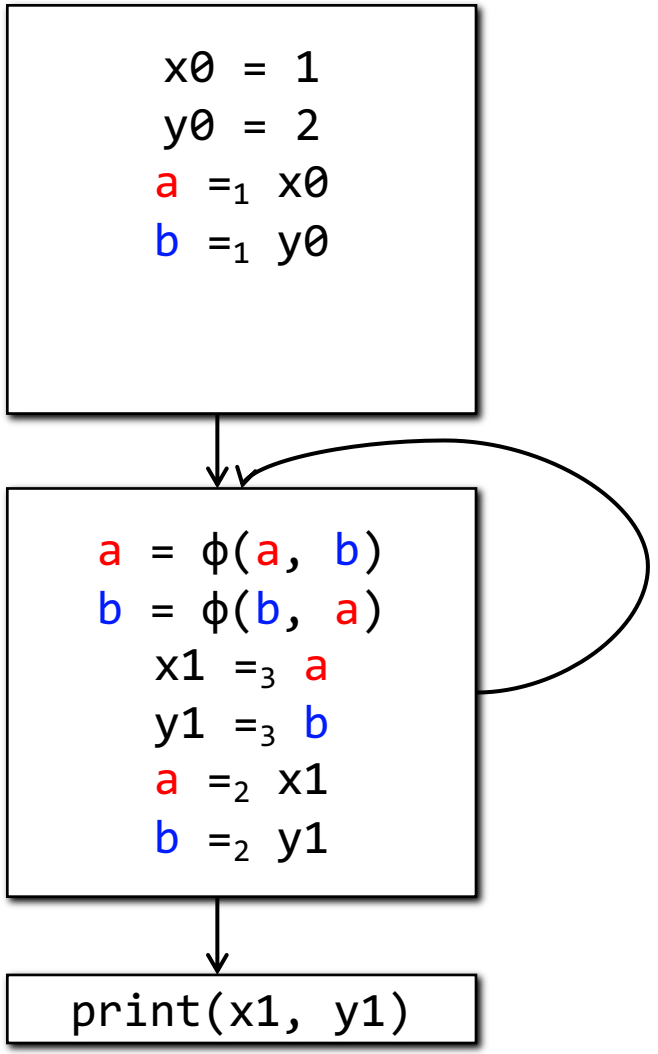

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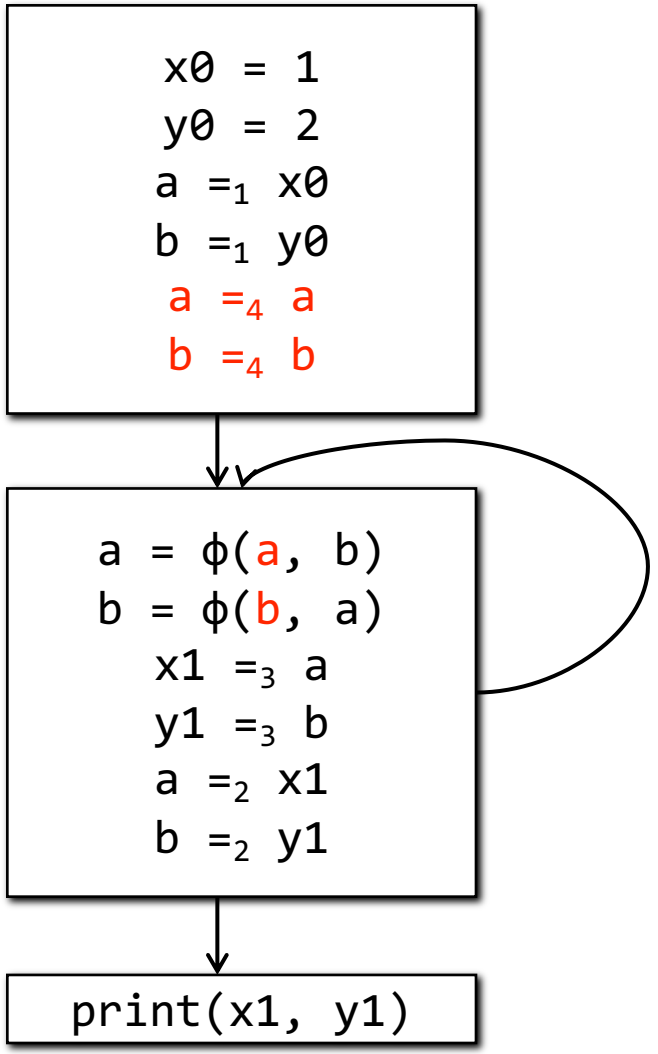

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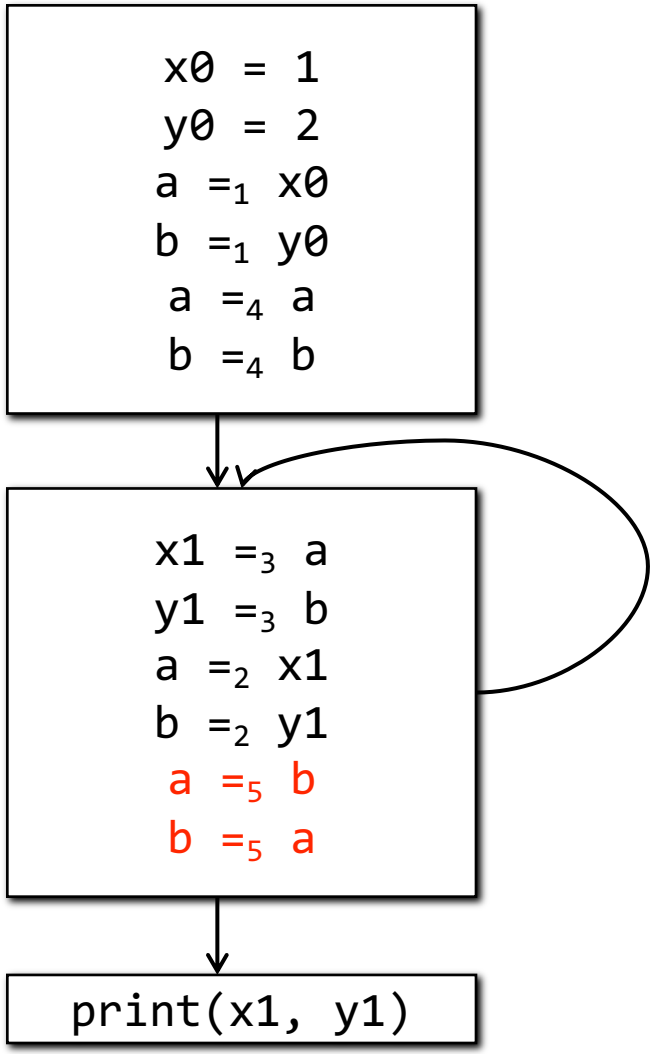

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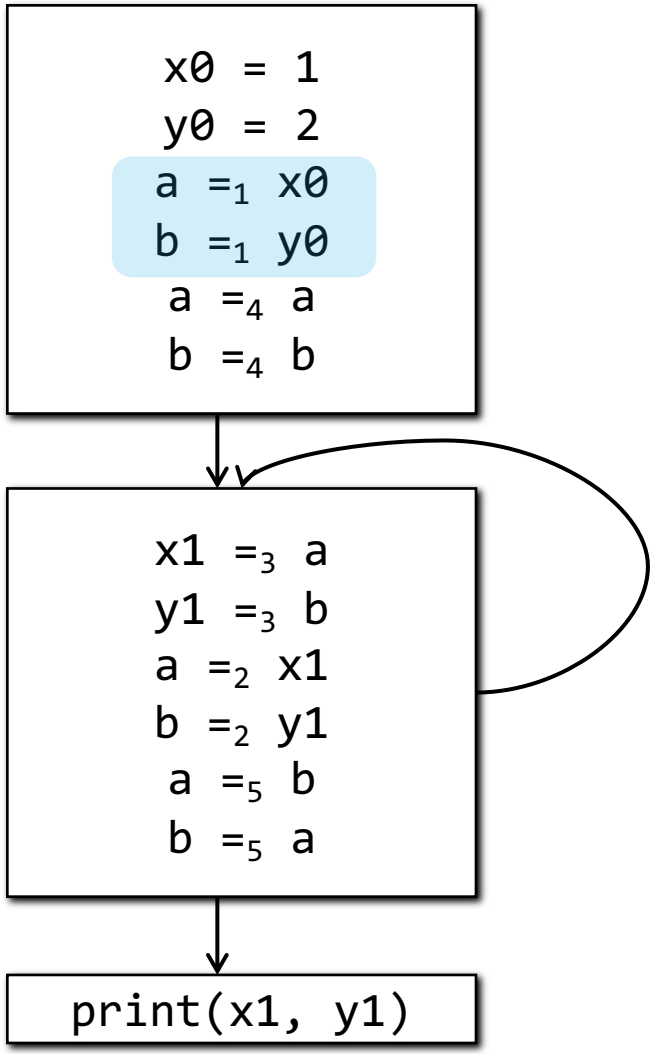

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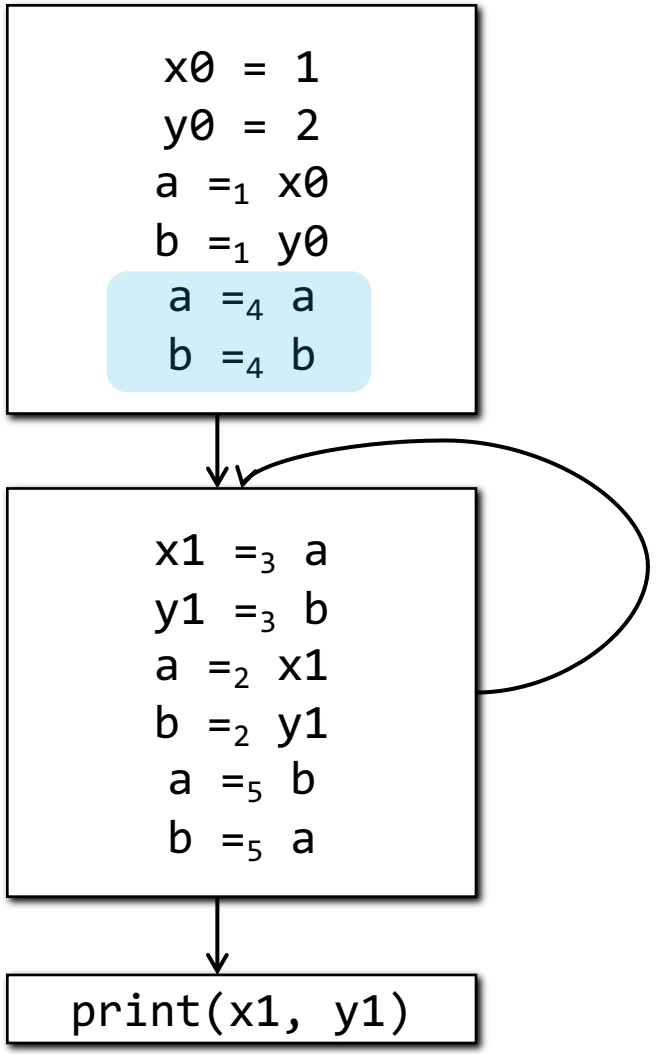

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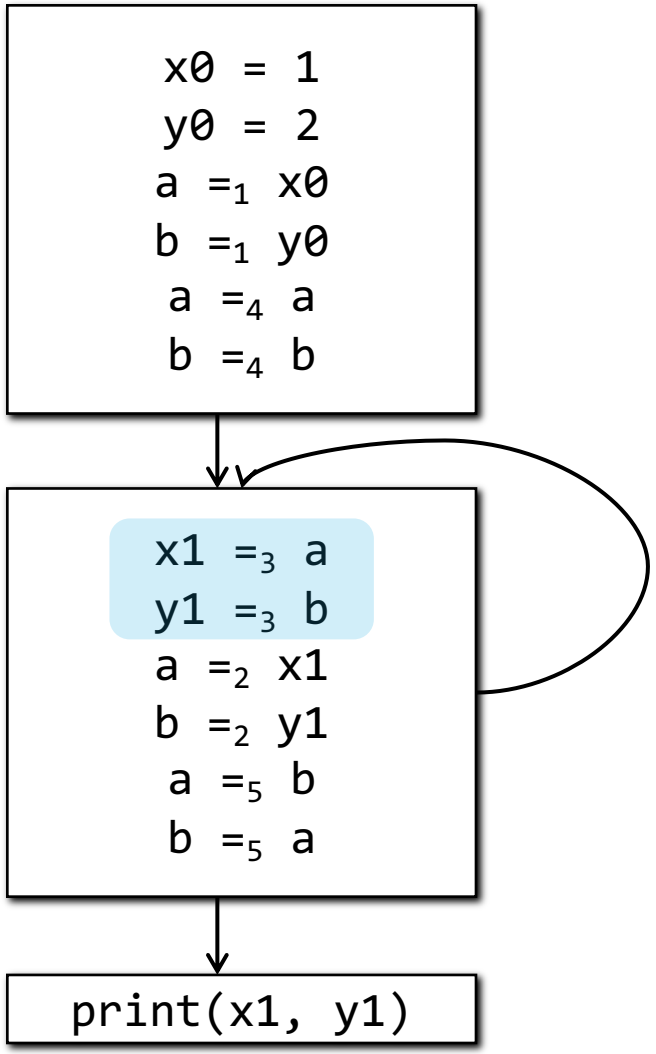

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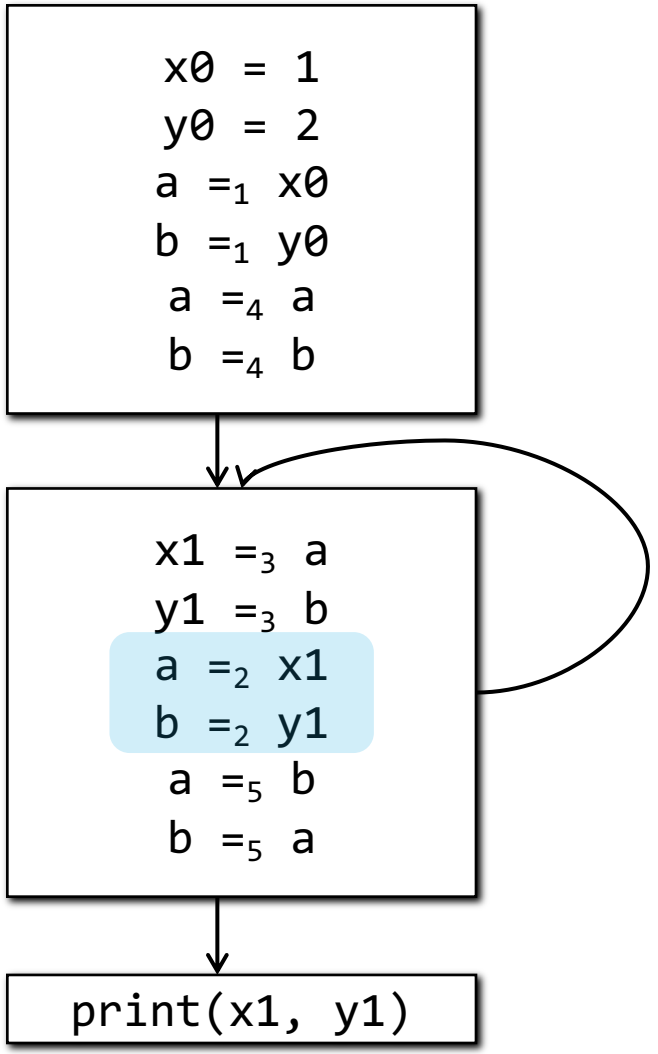

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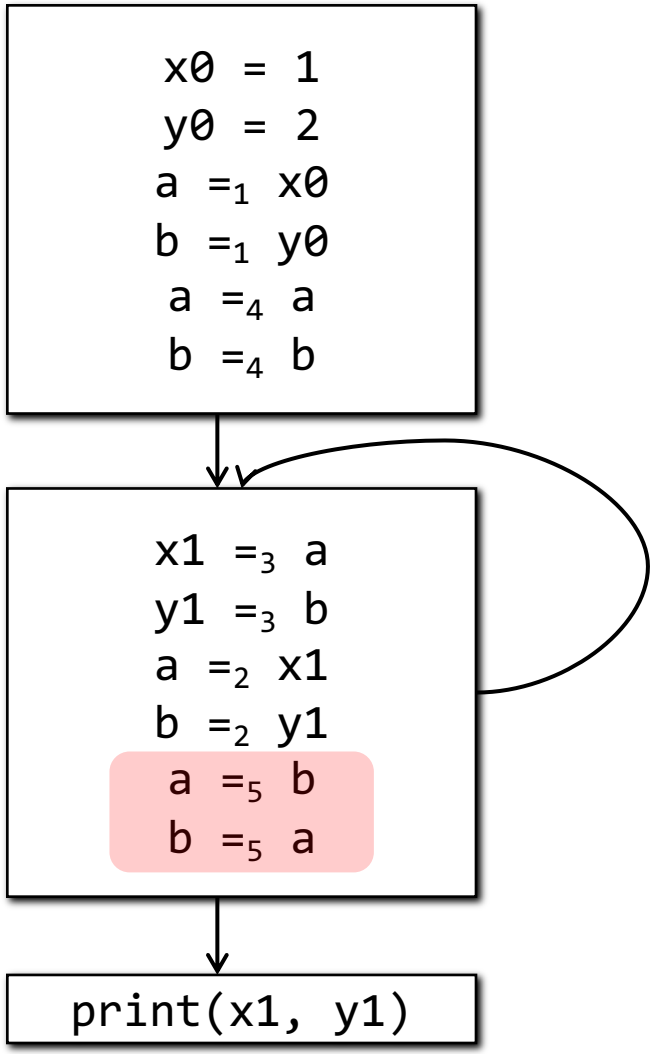

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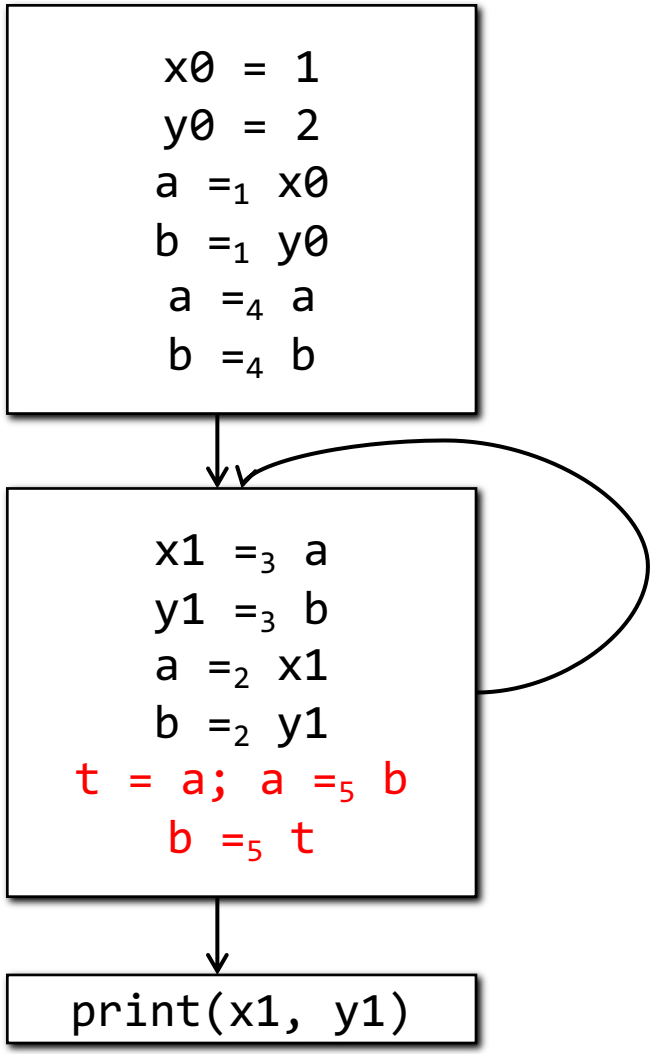

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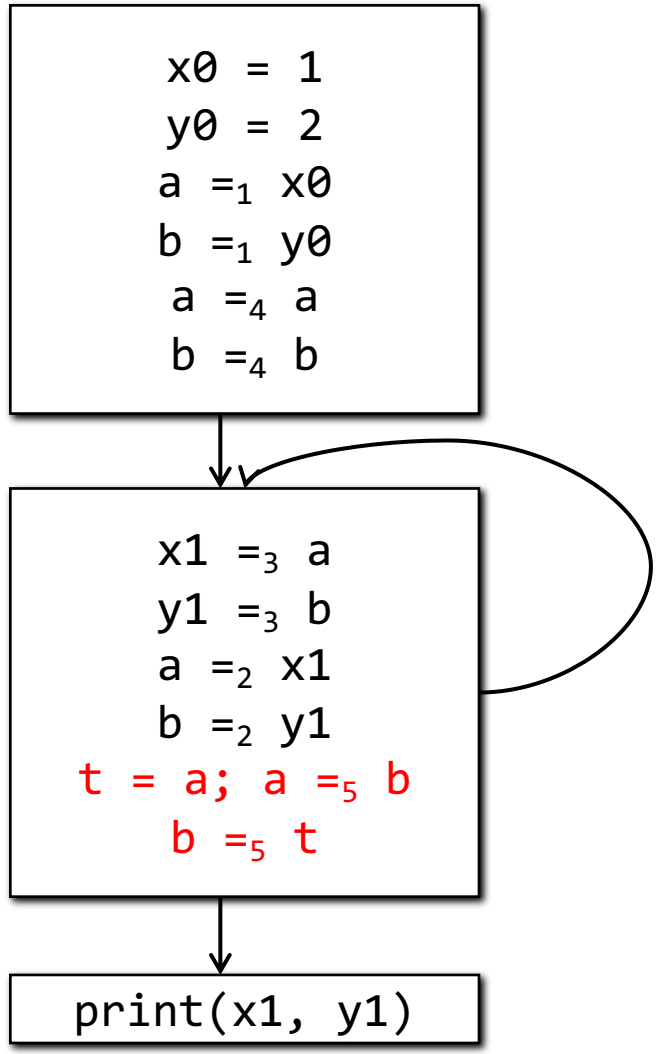

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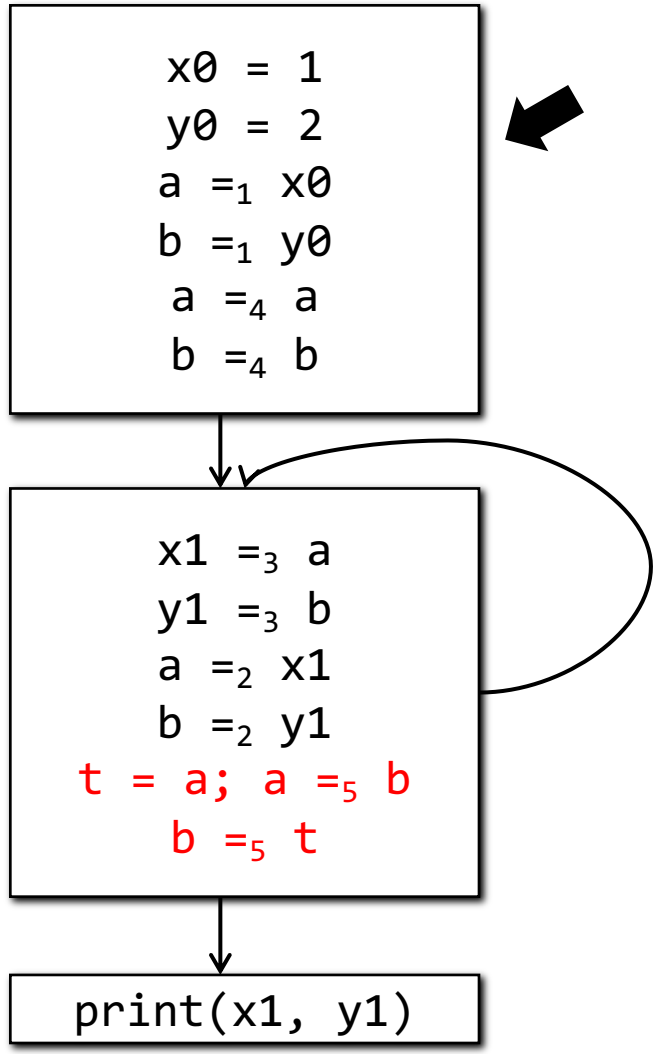

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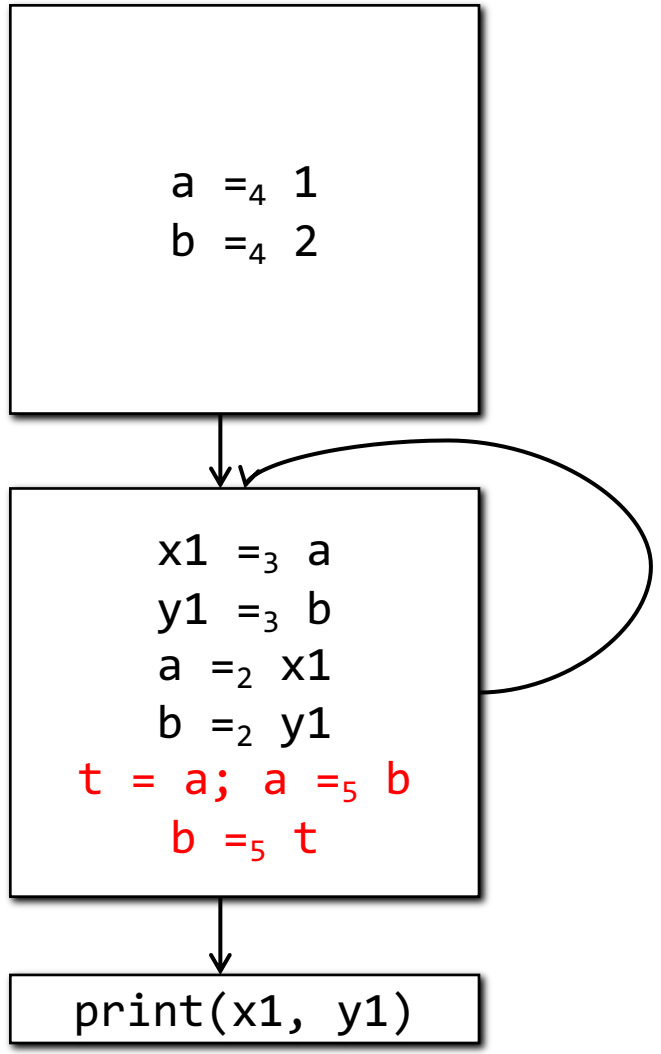

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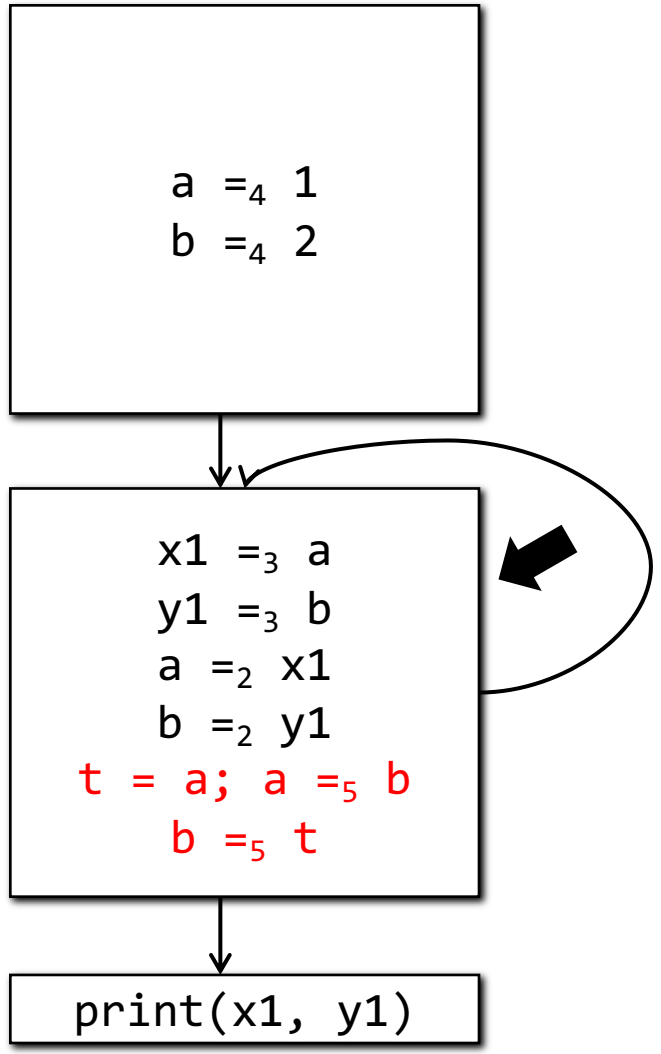

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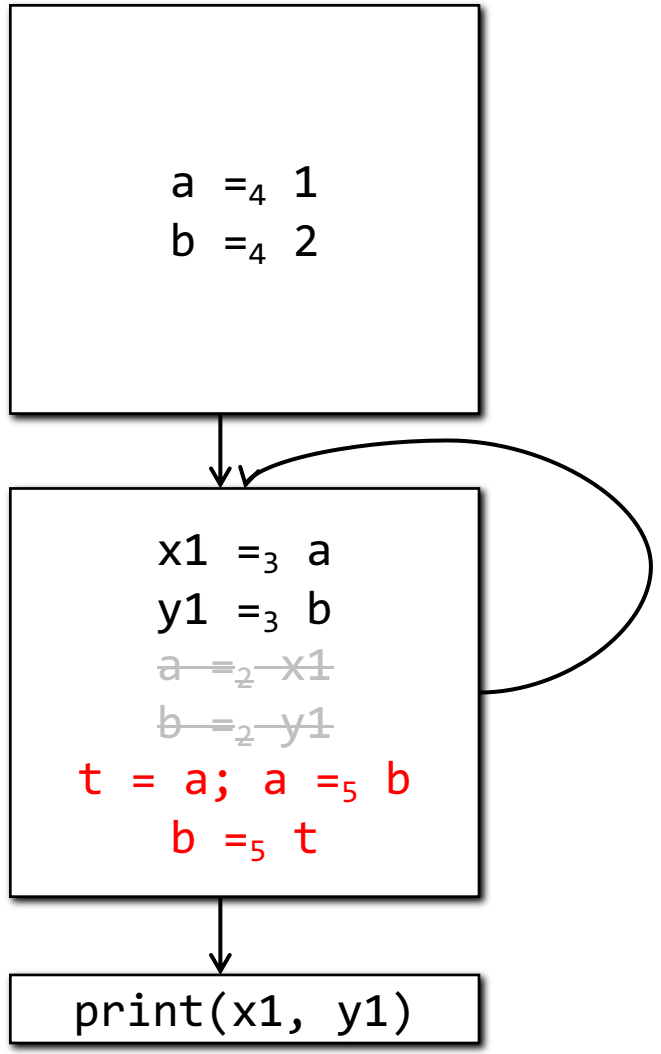
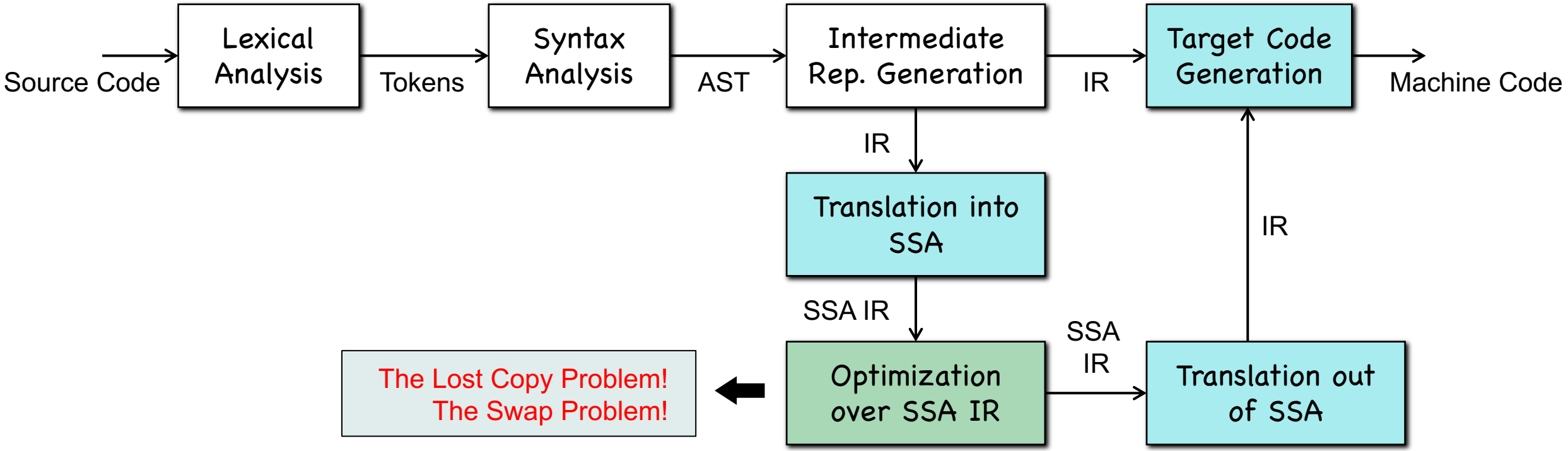


Summary


Source Code
Lexical Analysis
Tokens
Syntax Analysis
AST
Intermediate Rep. Generation
IR
Target Code Generation
Machine Code
IR
Translation into SSA
SSA IR
Optimization over SSA IR
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THANKS!